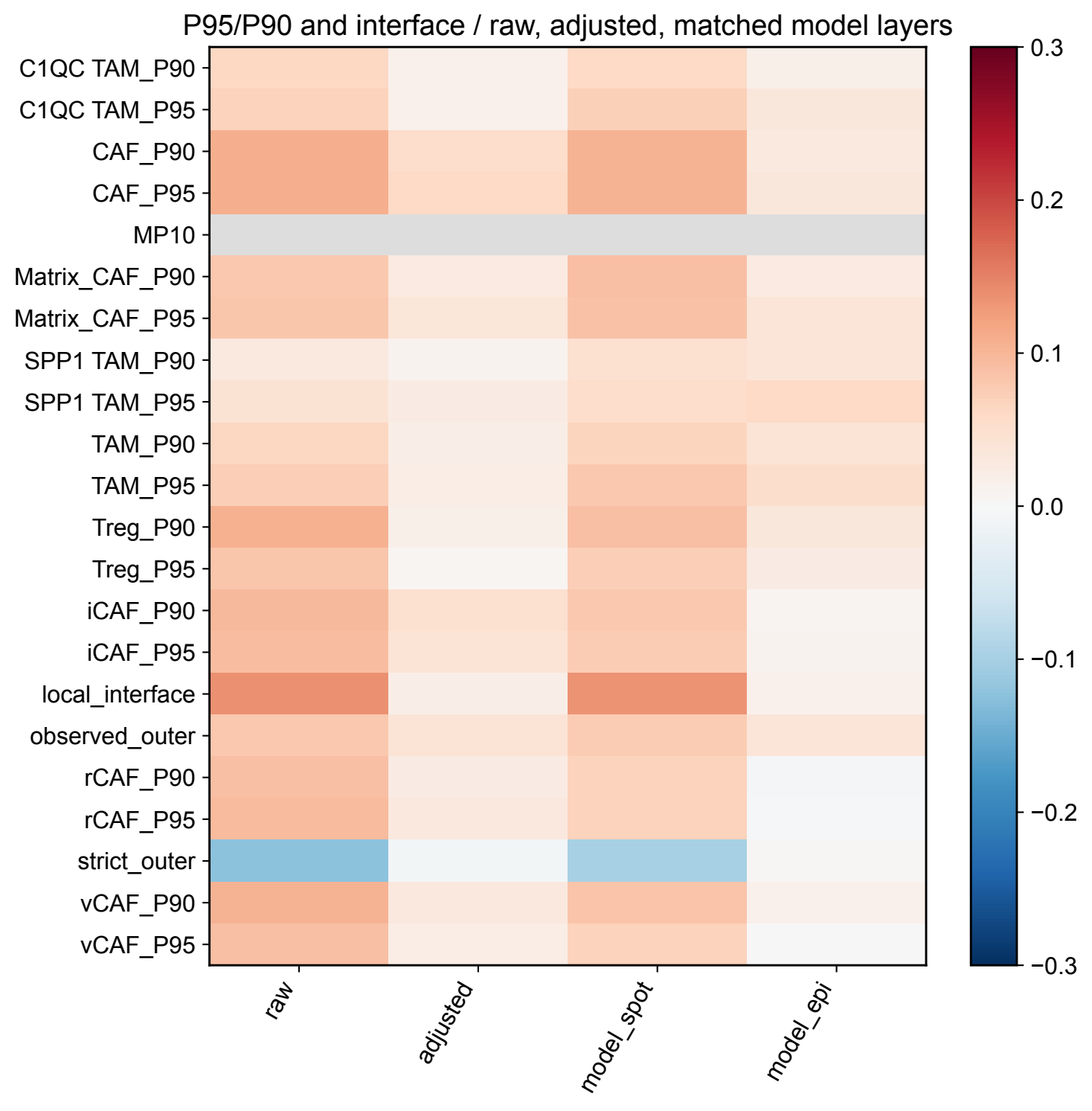
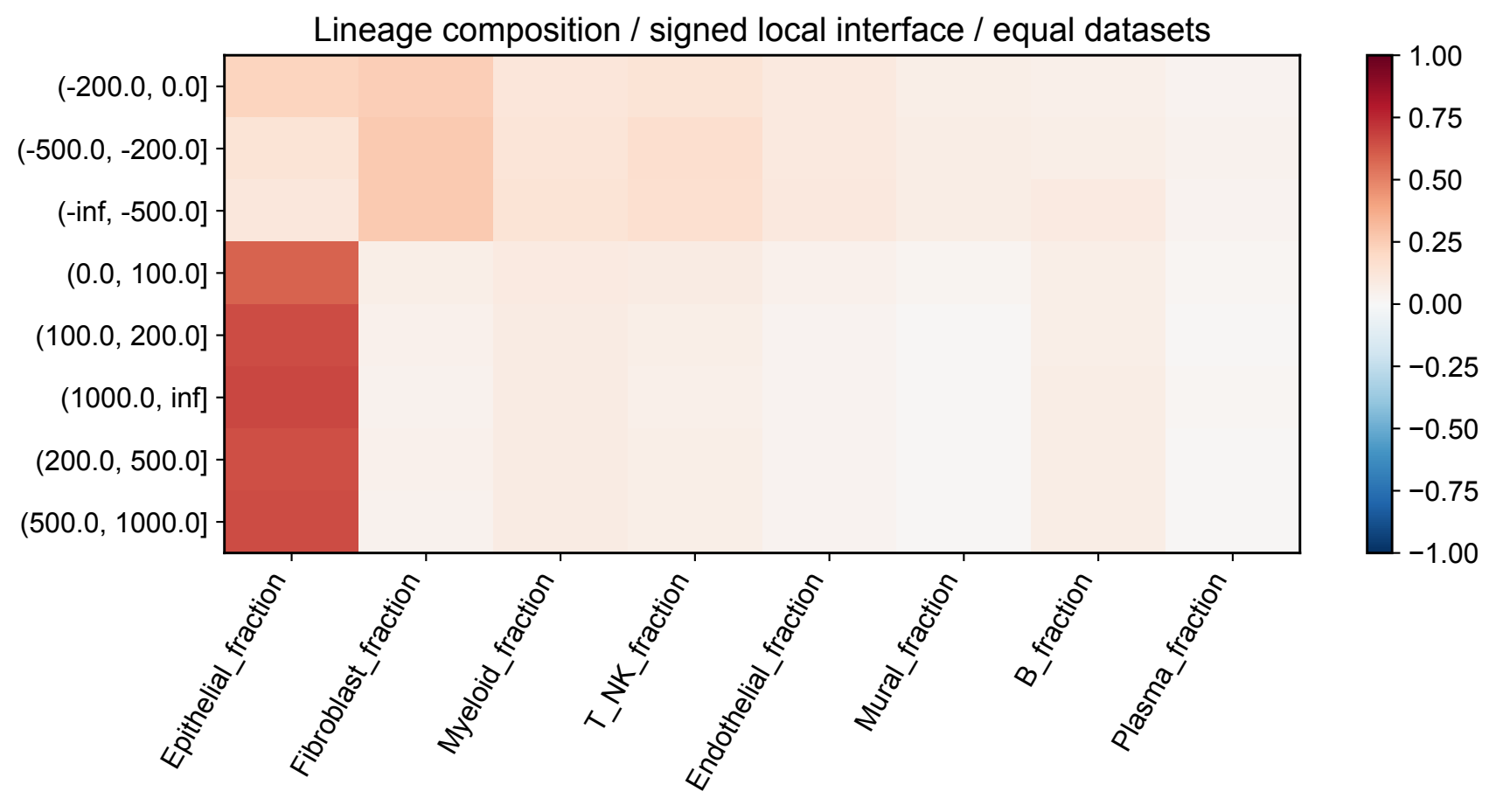

P95/P90 and interface / raw, adjusted, matched model layers
C1QC TAM_P90
C1QC TAM_P95
CAF_P90
CAF_P95
MP10
Matrix_CAF_P90
Matrix_CAF_P95
SPP1 TAM_P90
SPP1 TAM_P95
TAM_P90
TAM_P95
Treg_P90
Treg_P95
iCAF_P90
iCAF_P95
local_interface
observed_outer
rCAF_P90
rCAF_P95
strict_outer
vCAF_P90
vCAF_P95
raw
adjusted
model_spot
model_epi
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3

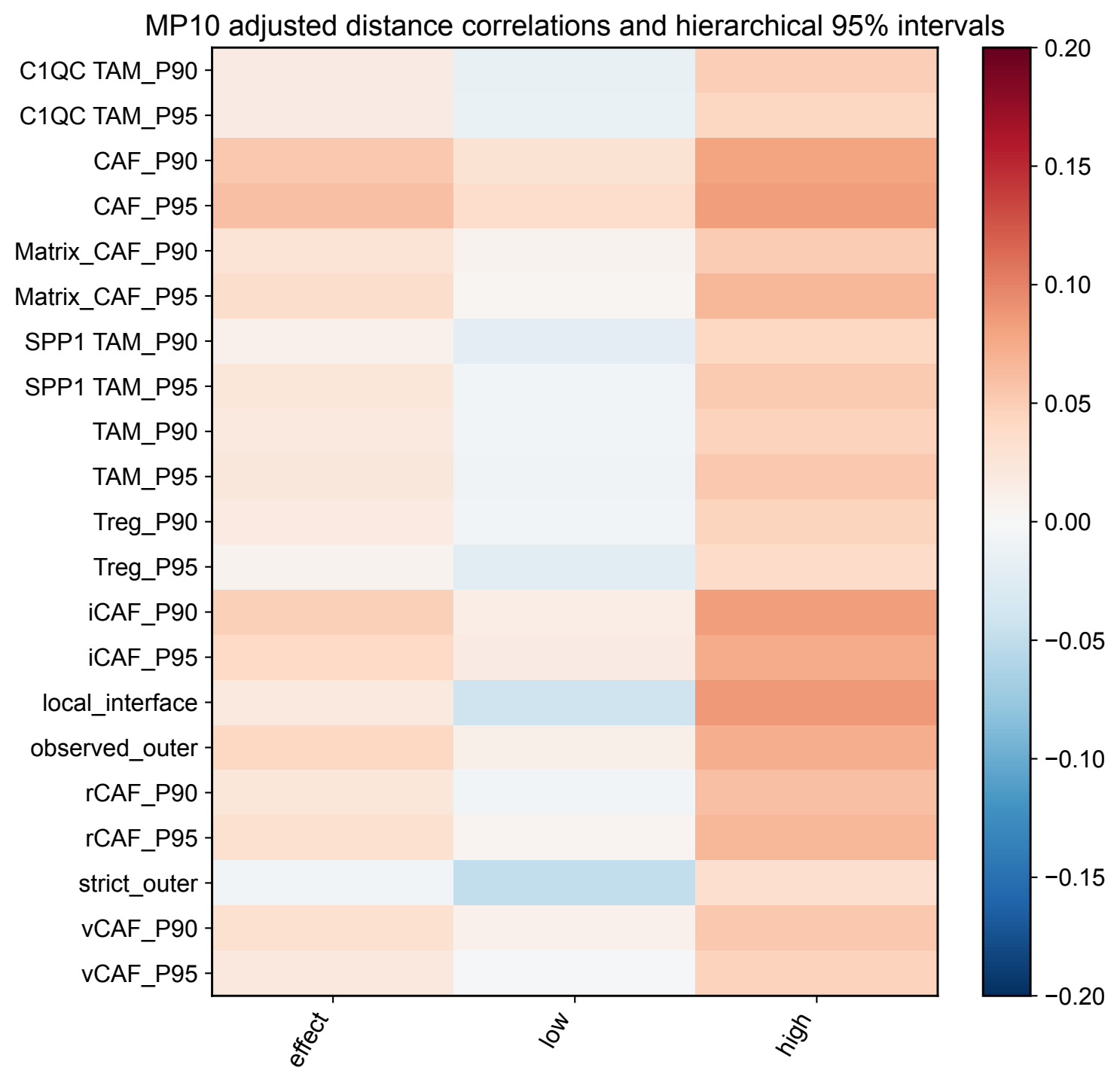

Lineage composition / signed local interface / equal datasets
(-200.0, 0.0]
(-500.0, -200.0]
(-inf, -500.0]
(0.0, 100.0]
(100.0, 200.0]
(1000.0, inf]
(200.0, 500.0]
(500.0, 1000.0]
Epithelial_fraction
Fibroblast_fraction
Myeloid_fraction
T_NK_fraction
Endothelial_fraction
Mural_fraction
B_fraction
Plasma_fraction
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

P95 target estimated abundance (log1p median)

CanDis_2026_Sample1
CanDis_2026_Sample10
CanDis_2026_Sample11
CanDis_2026_Sample12
CanDis_2026_Sample13
CanDis_2026_Sample2
CanDis_2026_Sample3
CanDis_2026_Sample4
CanDis_2026_Sample5
CanDis_2026_Sample6
CanDis_2026_Sample7
CanDis_2026_Sample8
CanDis_2026_Sample9
GSE171351_A1
GSE171351_B1
GSE171351_C1
GSE171351_D1
GSE238145_B10
GSE238145_B11
GSE238145_B13
GSE238145_B14
GSE238145_B15
GSE238145_B16
GSE238145_B17
GSE238145_B18
GSE238145_B5
GSE238145_B9
GSE285715_N1
GSE285715_N2
GSE285715_N3
GSE285715_T1
GSE285715_T2
GSE285715_T3
GSE319536_BC01
GSE319536_BC02
GSE319536_BC04
GSE319536_BC05
GSE319536_BC08
GSE319536_BC10
GSE319536_BC11
GSE319536_BC12
GSE319536_BC13
GSE319536_BC14
GSE319536_BC15
GSE319536_BC16
GSE319536_BC17
GSE319536_BC18
GSE319536_BC19
GSE319536_BC21
Inhousedata_T4384654
Inhousedata_T4398343
Inhousedata_T4413900
Inhousedata_T4463209
Inhousedata_T4465935
Inhousedata_T4469479
Inhousedata_T4471438
NPJ_sample1
NPJ_sample2
NPJ_sample3
NPJ_sample4
NPJ_sample5
NPJ_sample6
NPJ_sample7

C1QC TAM
CAF
Matrix_CAF
SPP1 TAM
TAM
Treg
iCAF
rCAF
vCAF

2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0

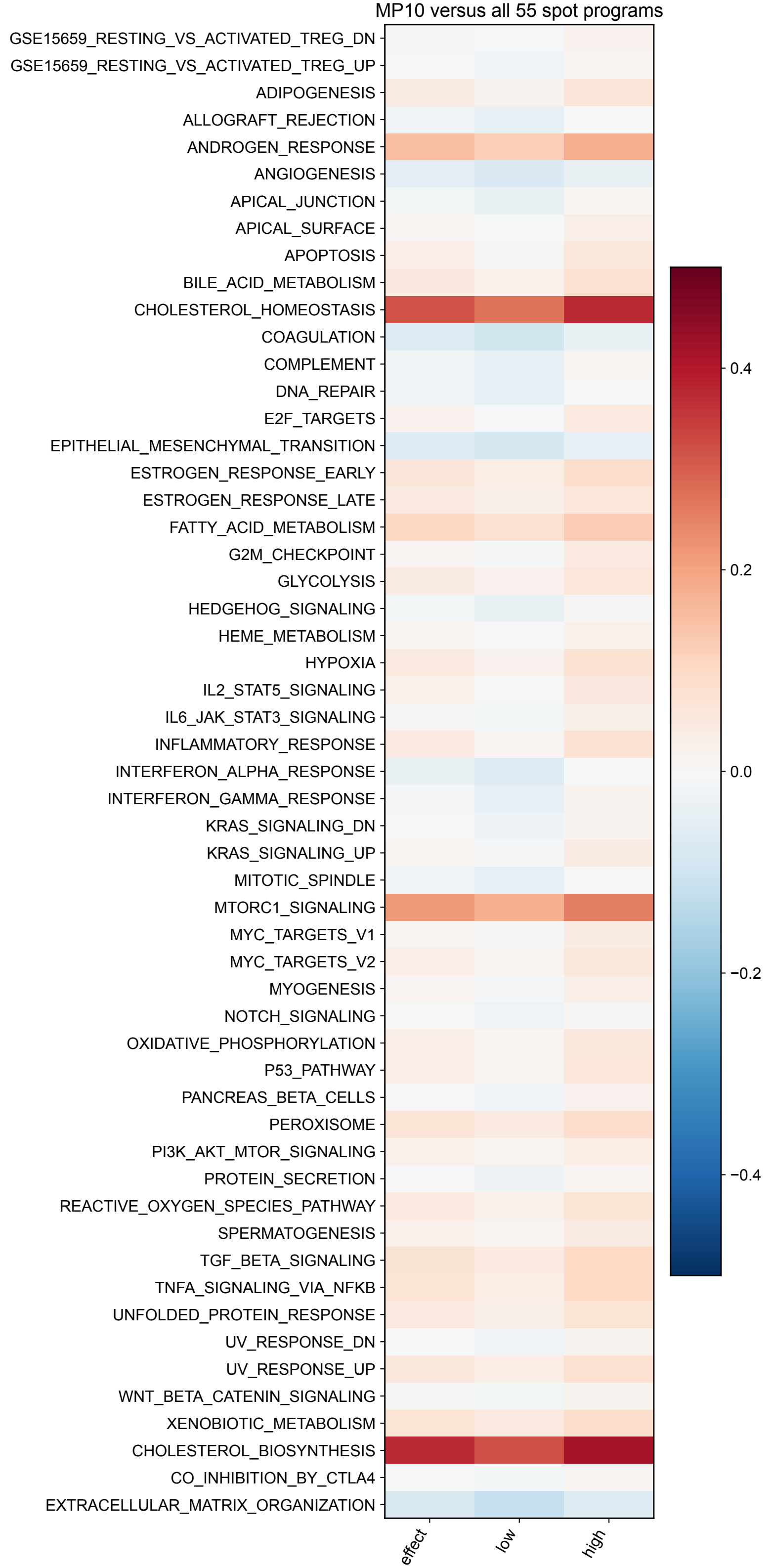

MP10 adjusted distance correlations and hierarchical 95% intervals
C1QC TAM_P90
C1QC TAM_P95
CAF_P90
CAF_P95
Matrix_CAF_P90
Matrix_CAF_P95
SPP1 TAM_P90
SPP1 TAM_P95
TAM_P90
TAM_P95
Treg_P90
Treg_P95
iCAF_P90
iCAF_P95
local_interface
observed_outer
rCAF_P90
rCAF_P95
strict_outer
vCAF_P90
vCAF_P95
effect
low
high
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20

P95 / adjusted effects / equal datasets of analysis units

GSE15659_RESTING_VS_ACTIVATED_TREG_DN
GSE15659_RESTING_VS_ACTIVATED_TREG_UP
ADIPOGENESIS
ALLOGRAFT_REJECTION
ANDROGEN_RESPONSE
ANGIOGENESIS
APICAL_JUNCTION
APICAL_SURFACE
APOPTOSIS
BILE_ACID_METABOLISM
CHOLESTEROL_HOMEOSTASIS
COAGULATION
COMPLEMENT
DNA_REPAIR
E2F_TARGETS
EPITHELIAL_MESENCHYMAL_TRANSITION
ESTROGEN_RESPONSE_EARLY
ESTROGEN_RESPONSE_LATE
FATTY_ACID_METABOLISM
G2M_CHECKPOINT
GLYCOLYSIS
HEDGEHOG_SIGNALING
HEME_METABOLISM
HYPOXIA
IL2_STAT5_SIGNALING
IL6_JAK_STAT3_SIGNALING
INFLAMMATORY_RESPONSE
INTERFERON_ALPHA_RESPONSE
INTERFERON_GAMMA_RESPONSE
KRAS_SIGNALING_DN
KRAS_SIGNALING_UP
MITOTIC_SPINDLE
MTORC1_SIGNALING
MYC_TARGETS_V1
MYC_TARGETS_V2
MYOGENESIS
NOTCH_SIGNALING
OXIDATIVE_PHOSPHORYLATION
P53_PATHWAY
PANCREAS_BETA_CELLS
PEROXISOME
PI3K_AKT_MTOR_SIGNALING
PROTEIN_SECRETION
REACTIVE_OXYGEN_SPECIES_PATHWAY
SPERMATOGENESIS
TGF_BETA_SIGNALING
TNFA_SIGNALING_VIA_NFKB
UNFOLDED_PROTEIN_RESPONSE
UV_RESPONSE_DN
UV_RESPONSE_UP
WNT_BETA_CATENIN_SIGNALING
XENOBIOTIC_METABOLISM
CHOLESTEROL_BIOSYNTHESIS
CO_INHIBITION_BY_CTLA4
EXTRACELLULAR_MATRIX_ORGANIZATION

C1QC TAM_P95
CAF_P95
Matrix_CAF_P95
SPP1 TAM_P95
TAM_P95
Treg_P95
iCAF_P95
rCAF_P95
vCAF_P95

0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3


MP10 versus all 55 spot programs
GSE15659_RESTING_VS_ACTIVATED_TREG_DN
GSE15659_RESTING_VS_ACTIVATED_TREG_UP
ADIPOGENESIS
ALLOGRAFT_REJECTION
ANDROGEN_RESPONSE
ANGIOGENESIS
APICAL_JUNCTION
APICAL_SURFACE
APOPTOSIS
BILE_ACID_METABOLISM
CHOLESTEROL_HOMEOSTASIS
COAGULATION
COMPLEMENT
DNA_REPAIR
E2F_TARGETS
EPITHELIAL_MESENCHYMAL_TRANSITION
ESTROGEN_RESPONSE_EARLY
ESTROGEN_RESPONSE_LATE
FATTY_ACID_METABOLISM
G2M_CHECKPOINT
GLYCOLYSIS
HEDGEHOG_SIGNALING
HEME_METABOLISM
HYPOXIA
IL2_STAT5_SIGNALING
IL6_JAK_STAT3_SIGNALING
INFLAMMATORY_RESPONSE
INTERFERON_ALPHA_RESPONSE
INTERFERON_GAMMA_RESPONSE
KRAS_SIGNALING_DN
KRAS_SIGNALING_UP
MITOTIC_SPINDLE
MTORC1_SIGNALING
MYC_TARGETS_V1
MYC_TARGETS_V2
MYOGENESIS
NOTCH_SIGNALING
OXIDATIVE_PHOSPHORYLATION
P53_PATHWAY
PANCREAS_BETA_CELLS
PEROXISOME
PI3K_AKT_MTOR_SIGNALING
PROTEIN_SECRETION
REACTIVE_OXYGEN_SPECIES_PATHWAY
SPERMATOGENESIS
TGF_BETA_SIGNALING
TNFA_SIGNALING_VIA_NFKB
UNFOLDED_PROTEIN_RESPONSE
UV_RESPONSE_DN
UV_RESPONSE_UP
WNT_BETA_CATENIN_SIGNALING
XENOBIOTIC_METABOLISM
CHOLESTEROL_BIOSYNTHESIS
CO_INHIBITION_BY_CTLA4
EXTRACELLULAR_MATRIX_ORGANIZATION
effect
low
high
0.4
0.2
0.0
−0.2
−0.4

P95 / adjusted effects / equal datasets of analysis units

GSE15659_RESTING_VS_ACTIVATED_TREG_DN
GSE15659_RESTING_VS_ACTIVATED_TREG_UP
ADIPOGENESIS
ALLOGRAFT_REJECTION
ANDROGEN_RESPONSE
ANGIOGENESIS
APICAL_JUNCTION
APICAL_SURFACE
APOPTOSIS
BILE_ACID_METABOLISM
CHOLESTEROL_HOMEOSTASIS
COAGULATION
COMPLEMENT
DNA_REPAIR
E2F_TARGETS
EPITHELIAL_MESENCHYMAL_TRANSITION
ESTROGEN_RESPONSE_EARLY
ESTROGEN_RESPONSE_LATE
FATTY_ACID_METABOLISM
G2M_CHECKPOINT
GLYCOLYSIS
HEDGEHOG_SIGNALING
HEME_METABOLISM
HYPOXIA
IL2_STAT5_SIGNALING
IL6_JAK_STAT3_SIGNALING
INFLAMMATORY_RESPONSE
INTERFERON_ALPHA_RESPONSE
INTERFERON_GAMMA_RESPONSE
KRAS_SIGNALING_DN
KRAS_SIGNALING_UP
MITOTIC_SPINDLE
MTORC1_SIGNALING
MYC_TARGETS_V1
MYC_TARGETS_V2
MYOGENESIS
NOTCH_SIGNALING
OXIDATIVE_PHOSPHORYLATION
P53_PATHWAY
PANCREAS_BETA_CELLS
PEROXISOME
PI3K_AKT_MTOR_SIGNALING
PROTEIN_SECRETION
REACTIVE_OXYGEN_SPECIES_PATHWAY
SPERMATOGENESIS
TGF_BETA_SIGNALING
TNFA_SIGNALING_VIA_NFKB
UNFOLDED_PROTEIN_RESPONSE
UV_RESPONSE_DN
UV_RESPONSE_UP
WNT_BETA_CATENIN_SIGNALING
XENOBIOTIC_METABOLISM
CHOLESTEROL_BIOSYNTHESIS
CO_INHIBITION_BY_CTLA4
EXTRACELLULAR_MATRIX_ORGANIZATION

C1QC TAM_P95
CAF_P95
Matrix_CAF_P95
SPP1 TAM_P95
TAM_P95
Treg_P95
iCAF_P95
rCAF_P95
vCAF_P95

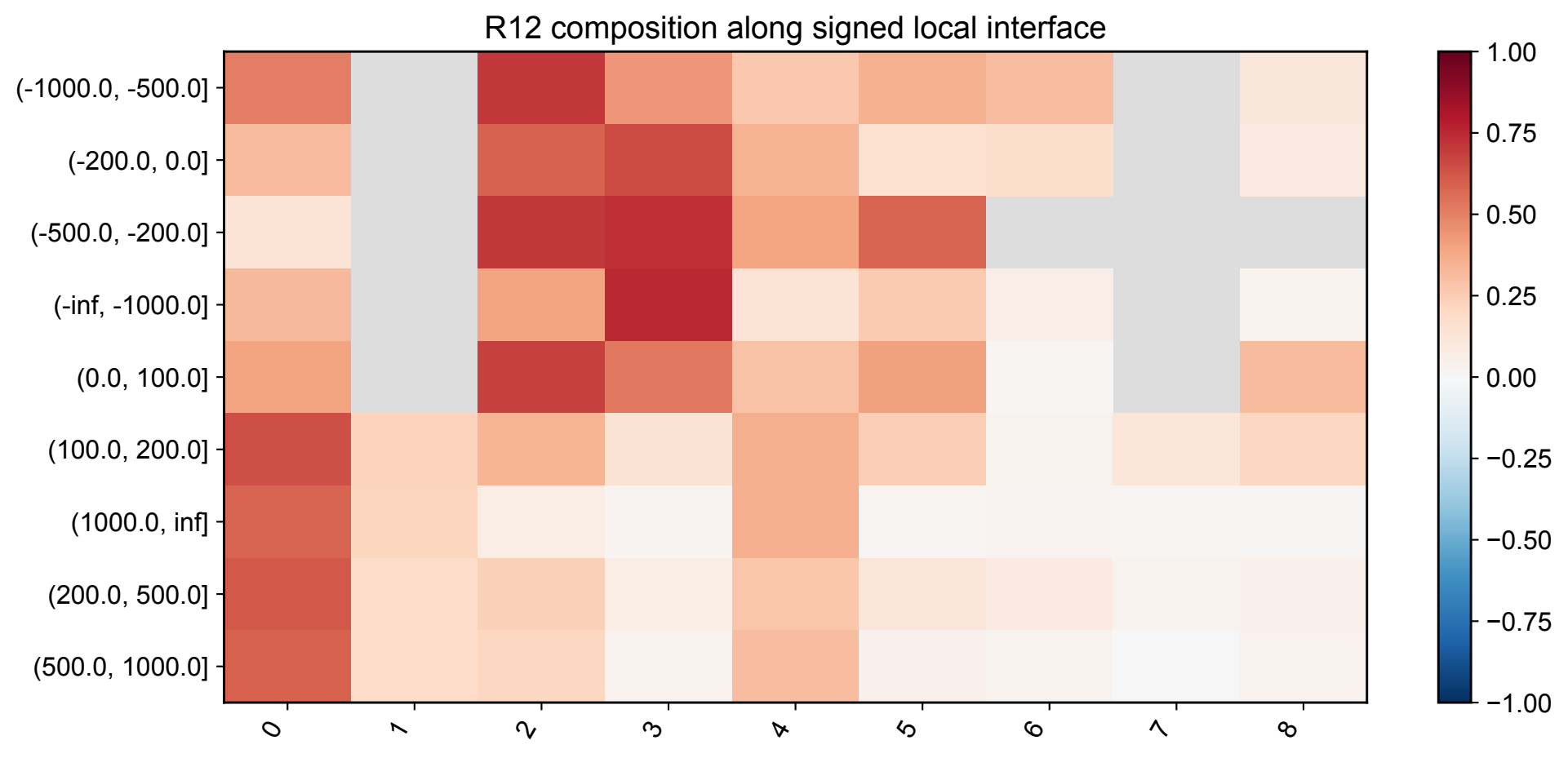
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08

Local interface and lesion-edge references

EpiMP10_Top20_AUCell
GSE15659_RESTING_VS_ACTIVATED_TREG_DN
GSE15659_RESTING_VS_ACTIVATED_TREG_UP
ADIPOGENESIS
ALLOGRAFT_REJECTION
ANDROGEN_RESPONSE
ANGIOGENESIS
APICAL_JUNCTION
APICAL_SURFACE
APOPTOSIS
BILE_ACID_METABOLISM
CHOLESTEROL_HOMEOSTASIS
COAGULATION
COMPLEMENT
DNA_REPAIR
E2F_TARGETS
EPITHELIAL_MESENCHYMAL_TRANSITION
ESTROGEN_RESPONSE_EARLY
ESTROGEN_RESPONSE_LATE
FATTY_ACID_METABOLISM
G2M_CHECKPOINT
GLYCOLYSIS
HEDGEHOG_SIGNALING
HEME_METABOLISM
HYPOXIA
IL2_STAT5_SIGNALING
IL6_JAK_STAT3_SIGNALING
INFLAMMATORY_RESPONSE
INTERFERON_ALPHA_RESPONSE
INTERFERON_GAMMA_RESPONSE
KRAS_SIGNALING_DN
KRAS_SIGNALING_UP
MITOTIC_SPINDLE
MTORC1_SIGNALING
MYC_TARGETS_V1
MYC_TARGETS_V2
MYOGENESIS
NOTCH_SIGNALING
OXIDATIVE_PHOSPHORYLATION
P53_PATHWAY
PANCREAS_BETA_CELLS
PEROXISOME
PI3K_AKT_MTOR_SIGNALING
PROTEIN_SECRETION
REACTIVE_OXYGEN_SPECIES_PATHWAY
SPERMATOGENESIS
TGF_BETA_SIGNALING
TNFA_SIGNALING_VIA_NFKB
UNFOLDED_PROTEIN_RESPONSE
UV_RESPONSE_DN
UV_RESPONSE_UP
WNT_BETA_CATENIN_SIGNALING
XENOBIOTIC_METABOLISM
CHOLESTEROL_BIOSYNTHESIS
CO_INHIBITION_BY_CTLA4
EXTRACELLULAR_MATRIX_ORGANIZATION

local_interface
observed_outer
strict_outer

0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3


R12 composition along signed local interface
(-1000.0, -500.0]
(-200.0, 0.0]
(-500.0, -200.0]
(-inf, -1000.0]
(0.0, 100.0]
(100.0, 200.0]
(1000.0, inf]
(200.0, 500.0]
(500.0, 1000.0]
0
1
2
3
4
5
6
7
8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

Within-niche mean interaction coefficient (descriptive)

accepted_label / 0
accepted_label / 1
accepted_label / 2
accepted_label / 3
accepted_label / 4
accepted_label / 5
accepted_label / 6
accepted_label / 7
accepted_label / 8
local_niche_id / CanDis__UNVERIFIED / CanDis__UNVERIFIED_N01
local_niche_id / CanDis__UNVERIFIED / CanDis__UNVERIFIED_N02
local_niche_id / CanDis__UNVERIFIED / CanDis__UNVERIFIED_N03
local_niche_id / CanDis__UNVERIFIED / CanDis__UNVERIFIED_N04
local_niche_id / CanDis__UNVERIFIED / CanDis__UNVERIFIED_N05
local_niche_id / CanDis__UNVERIFIED / CanDis__UNVERIFIED_N06
local_niche_id / CanDis__UNVERIFIED / CanDis__UNVERIFIED_N07
local_niche_id / GSE171351 / GSE171351_N01
local_niche_id / GSE171351 / GSE171351_N03
local_niche_id / GSE171351 / GSE171351_N04
local_niche_id / GSE171351 / GSE171351_N05
local_niche_id / GSE238145 / GSE238145_N02
local_niche_id / GSE238145 / GSE238145_N03
local_niche_id / GSE238145 / GSE238145_N04
local_niche_id / GSE238145 / GSE238145_N05
local_niche_id / GSE238145 / GSE238145_N06
local_niche_id / GSE238145 / GSE238145_N09
local_niche_id / GSE238145 / GSE238145_N10
local_niche_id / GSE238145 / GSE238145_N12
local_niche_id / GSE238145 / GSE238145_N14
local_niche_id / GSE238145 / GSE238145_N17
local_niche_id / GSE238145 / GSE238145_N18
local_niche_id / GSE285715 / GSE285715_N01
local_niche_id / GSE319536 / GSE319536_N02
local_niche_id / GSE319536 / GSE319536_N03
local_niche_id / GSE319536 / GSE319536_N04
local_niche_id / Inhousedata__UNVERIFIED / Inhousedata__UNVERIFIED_N01
local_niche_id / Inhousedata__UNVERIFIED / Inhousedata__UNVERIFIED_N03
local_niche_id / Inhousedata__UNVERIFIED / Inhousedata__UNVERIFIED_N04
local_niche_id / NPJ__UNVERIFIED / NPJ__UNVERIFIED_N01
local_niche_id / NPJ__UNVERIFIED / NPJ__UNVERIFIED_N03
local_niche_id / NPJ__UNVERIFIED / NPJ__UNVERIFIED_N04

C1QC TAM
Matrix_CAF
SPP1 TAM
Treg
iCAF
local_interface
rCAF
vCAF

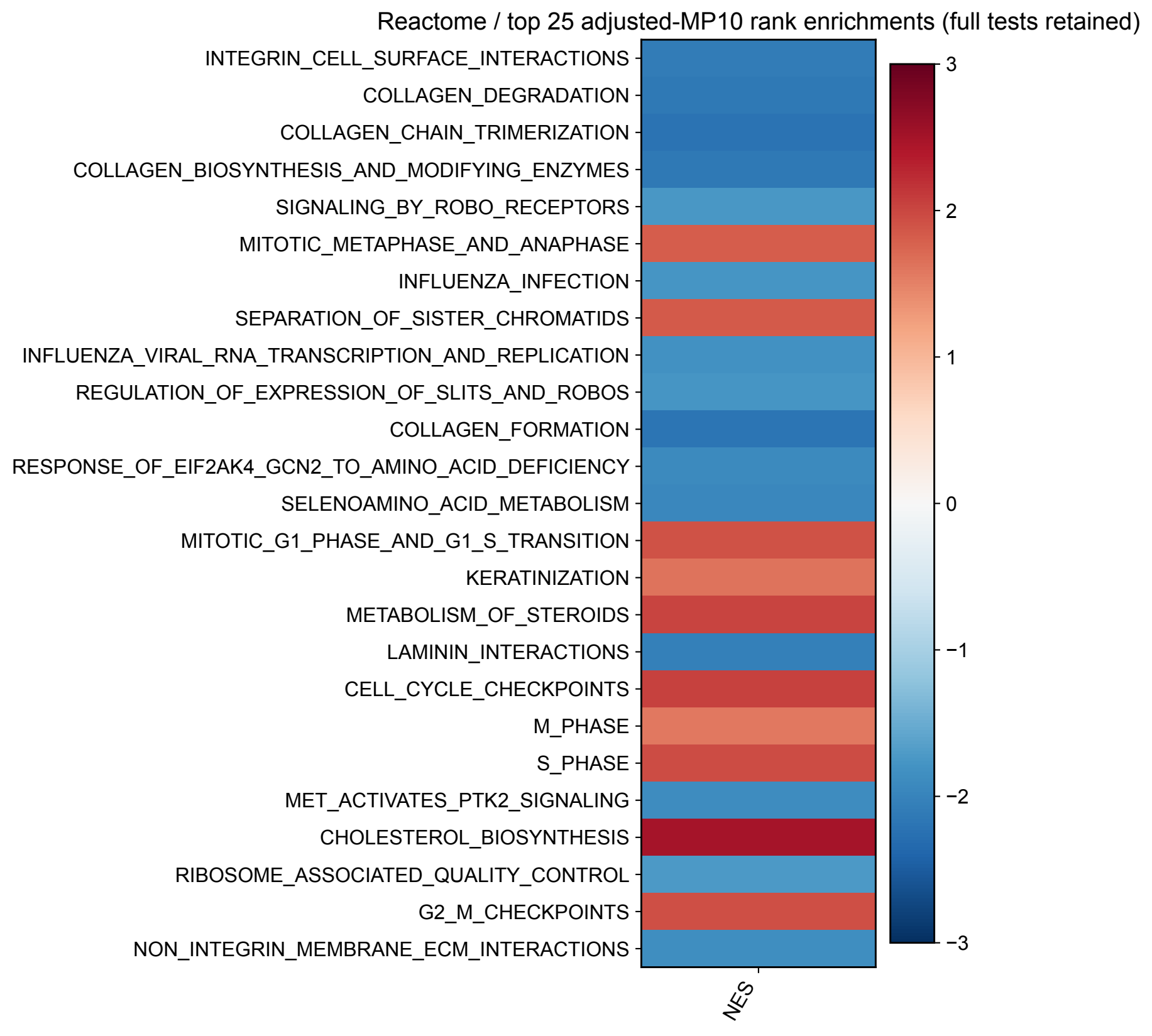
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100

Single-cell programs / median of sample medians

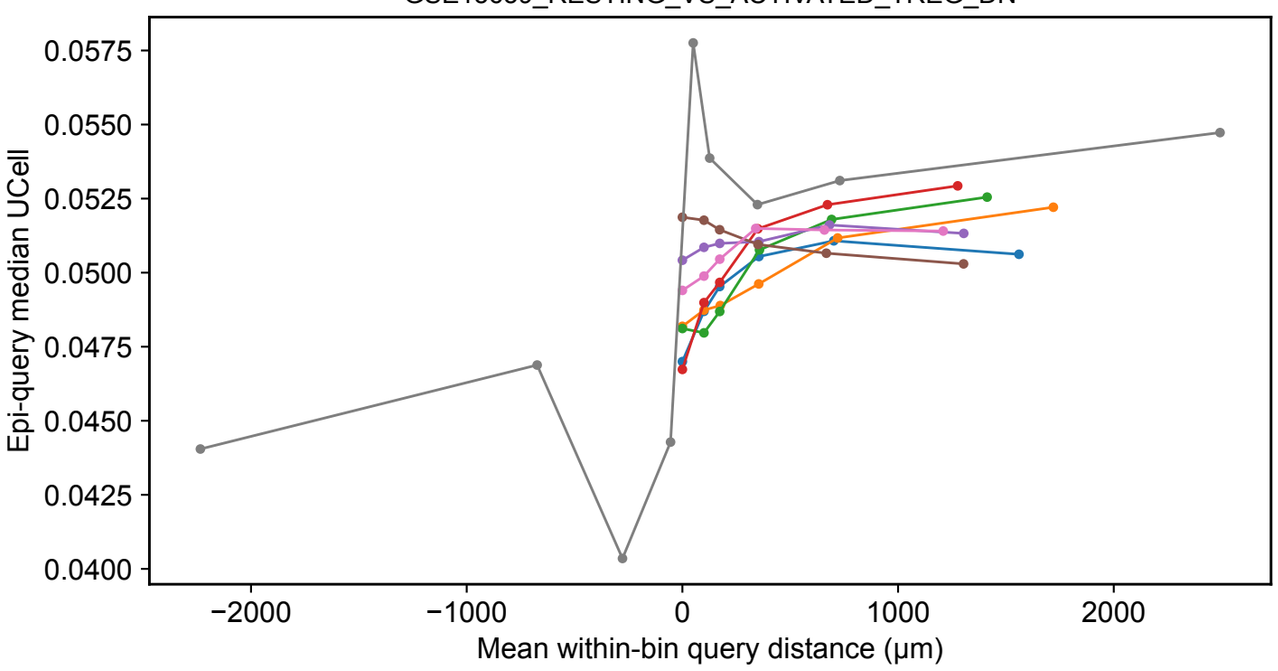
GSE15659_RESTING_VS_ACTIVATED_TREG_DN
GSE15659_RESTING_VS_ACTIVATED_TREG_UP
ADIPOGENESIS
ALLOGRAFT_REJECTION
ANDROGEN_RESPONSE
ANGIOGENESIS
APICAL_JUNCTION
APICAL_SURFACE
APOPTOSIS
BILE_ACID_METABOLISM
CHOLESTEROL_HOMEOSTASIS
COAGULATION
COMPLEMENT
DNA_REPAIR
E2F_TARGETS
EPITHELIAL_MESENCHYMAL_TRANSITION
ESTROGEN_RESPONSE_EARLY
ESTROGEN_RESPONSE_LATE
FATTY_ACID_METABOLISM
G2M_CHECKPOINT
GLYCOLYSIS
HEDGEHOG_SIGNALING
HEME_METABOLISM
HYPOXIA
IL2_STAT5_SIGNALING
IL6_JAK_STAT3_SIGNALING
INFLAMMATORY_RESPONSE
INTERFERON_ALPHA_RESPONSE
INTERFERON_GAMMA_RESPONSE
KRAS_SIGNALING_DN
KRAS_SIGNALING_UP
MITOTIC_SPINDLE
MTORC1_SIGNALING
MYC_TARGETS_V1
MYC_TARGETS_V2
MYOGENESIS
NOTCH_SIGNALING
OXIDATIVE_PHOSPHORYLATION
P53_PATHWAY
PANCREAS_BETA_CELLS
PEROXISOME
PI3K_AKT_MTOR_SIGNALING
PROTEIN_SECRETION
REACTIVE_OXYGEN_SPECIES_PATHWAY
SPERMATOGENESIS
TGF_BETA_SIGNALING
TNFA_SIGNALING_VIA_NFKB
UNFOLDED_PROTEIN_RESPONSE
UV_RESPONSE_DN
UV_RESPONSE_UP
WNT_BETA_CATENIN_SIGNALING
XENOBIOTIC_METABOLISM
CHOLESTEROL_BIOSYNTHESIS
CO_INHIBITION_BY_CTLA4
EXTRACELLULAR_MATRIX_ORGANIZATION

C1QC TAM
Epithelial
Matrix_CAF
SPP1 TAM
Treg
iCAF
rCAF
vCAF

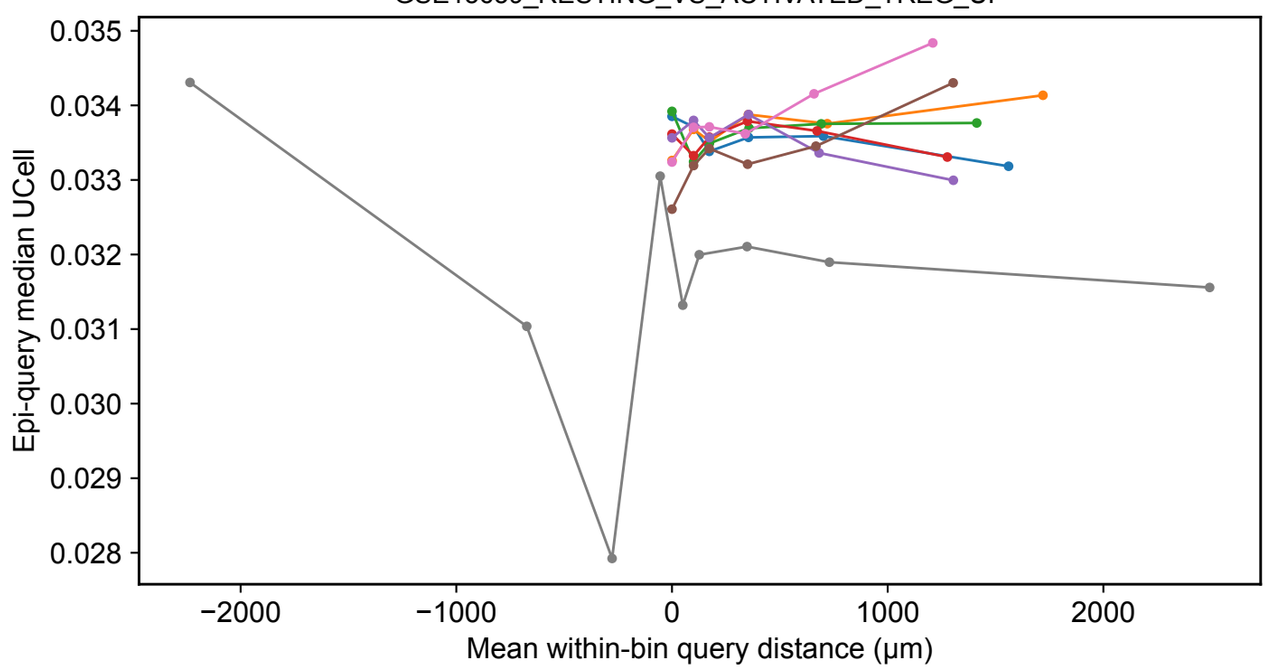
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3


Reactome / top 25 adjusted-MP10 rank enrichments (full tests retained)
INTEGRIN_CELL_SURFACE_INTERACTIONS
COLLAGEN_DEGRADATION
COLLAGEN_CHAIN_TRIMERIZATION
COLLAGEN_BIOSYNTHESIS_AND_MODIFYING_ENZYMES
SIGNALING_BY_ROBO_RECEPTORS
MITOTIC_METAPHASE_AND_ANAPHASE
INFLUENZA_INFECTION
SEPARATION_OF_SISTER_CHROMATIDS
INFLUENZA_VIRAL_RNA_TRANSCRIPTION_AND_REPLICATION
REGULATION_OF_EXPRESSION_OF_SLITS_AND_ROBOS
COLLAGEN_FORMATION
RESPONSE_OF_EIF2AK4_GCN2_TO_AMINO_ACID_DEFICIENCY
SELENOAMINO_ACID_METABOLISM
MITOTIC_G1_PHASE_AND_G1_S_TRANSITION
KERATINIZATION
METABOLISM_OF_STEROIDS
LAMININ_INTERACTIONS
CELL_CYCLE_CHECKPOINTS
M_PHASE
S_PHASE
MET_ACTIVATES_PTK2_SIGNALING
CHOLESTEROL_BIOSYNTHESIS
RIBOSOME_ASSOCIATED_QUALITY_CONTROL
G2_M_CHECKPOINTS
NON_INTEGRIN_MEMBRANE_ECM_INTERACTIONS
NES
3
2
1
0
−1
−2
−3

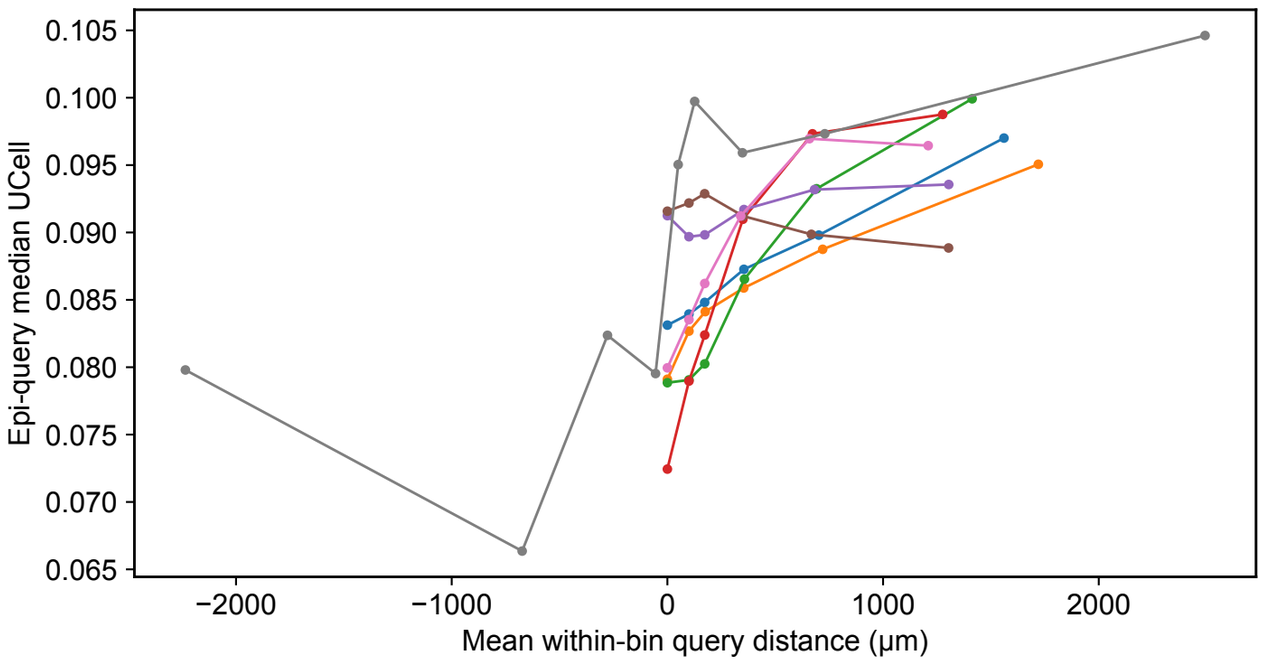
GSE15659_RESTING_VS_ACTIVATED_TREG_DN



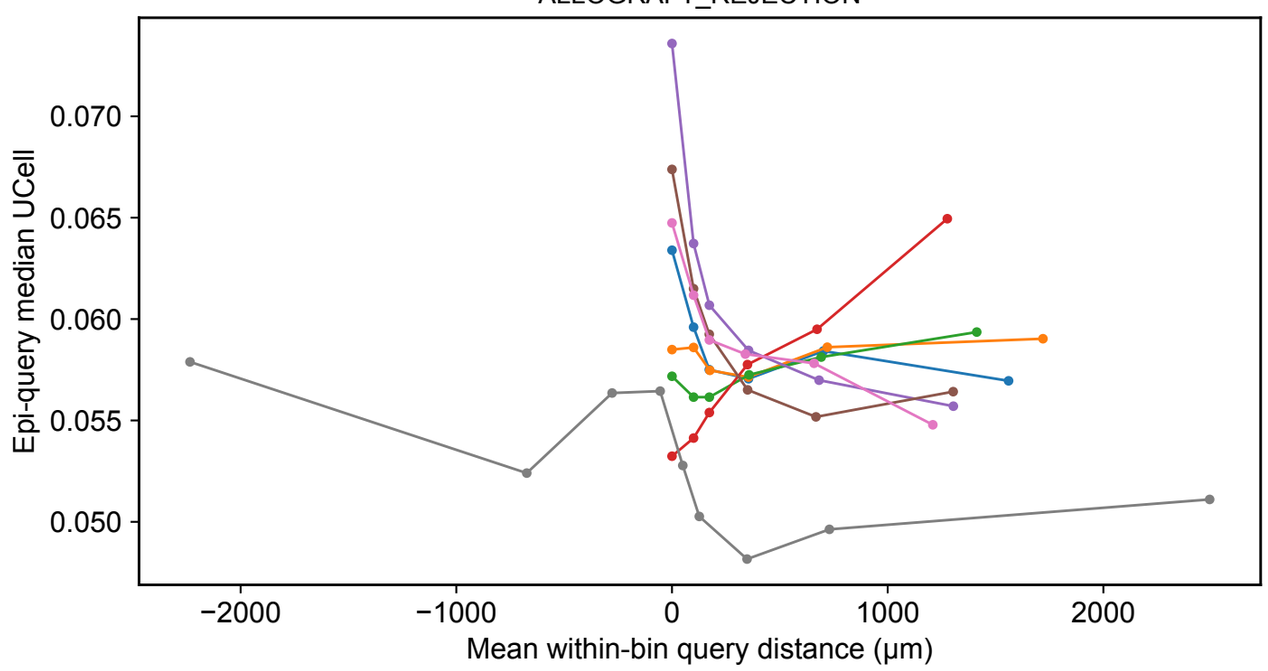
GSE15659_RESTING_VS_ACTIVATED_TREG_UP



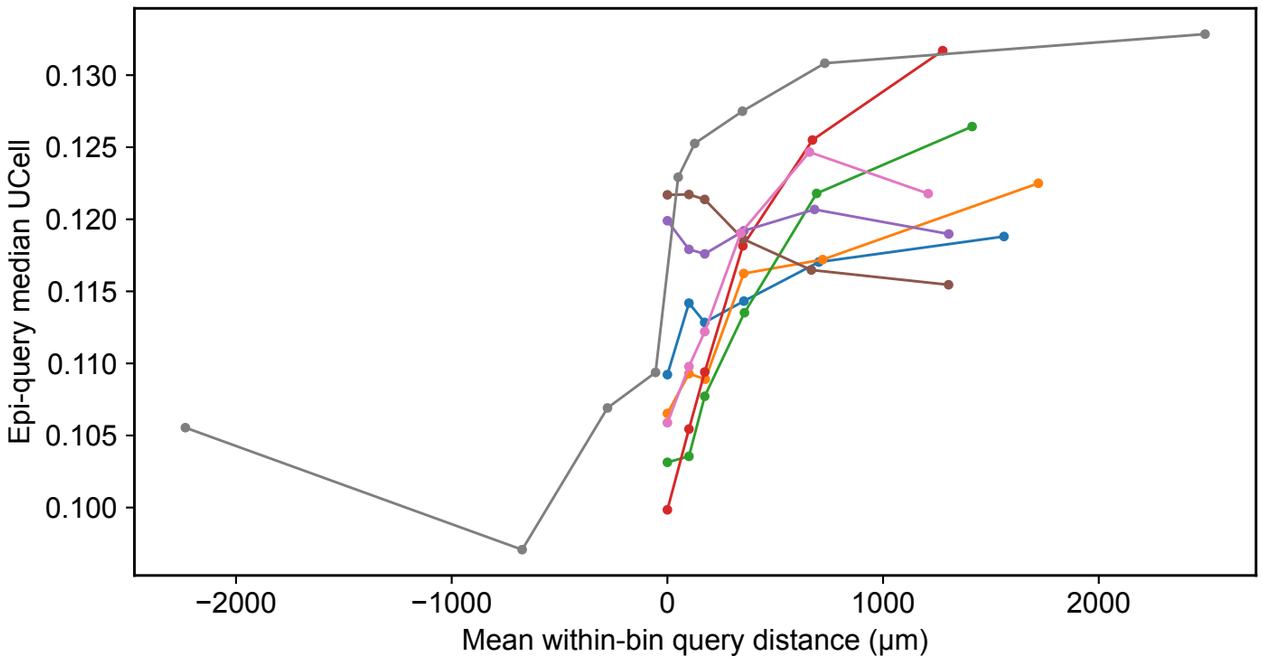
ADIPOGENESIS



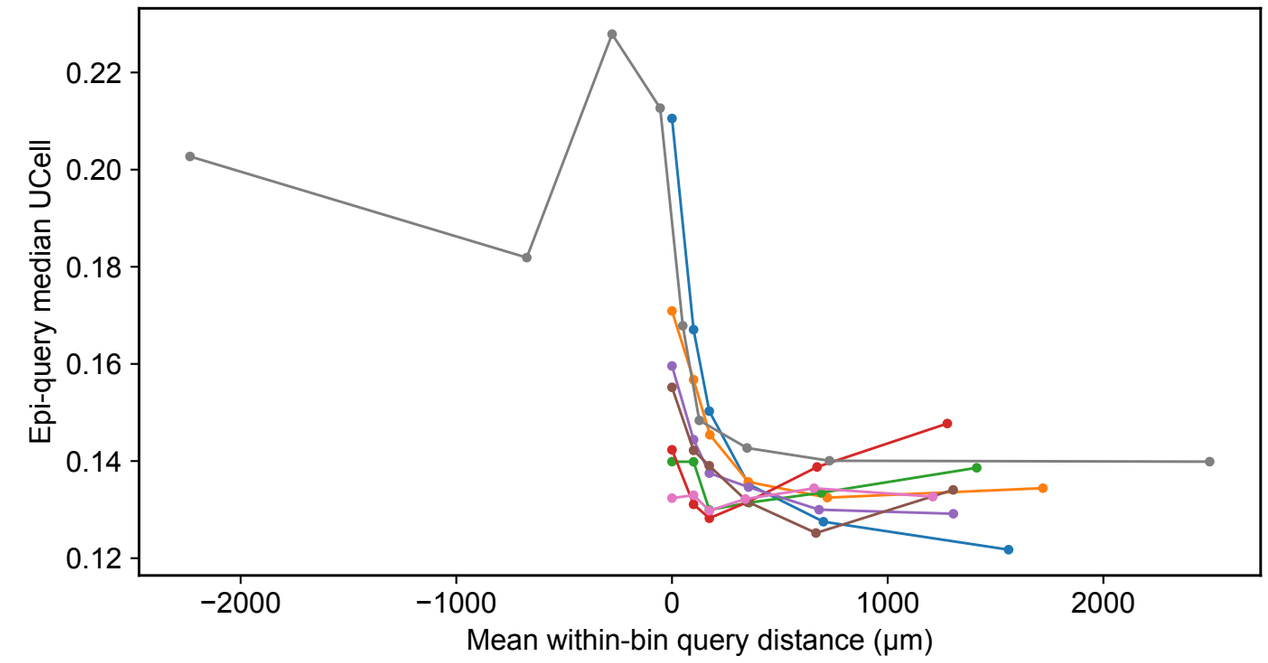
ALLOGRAFT_REJECTION



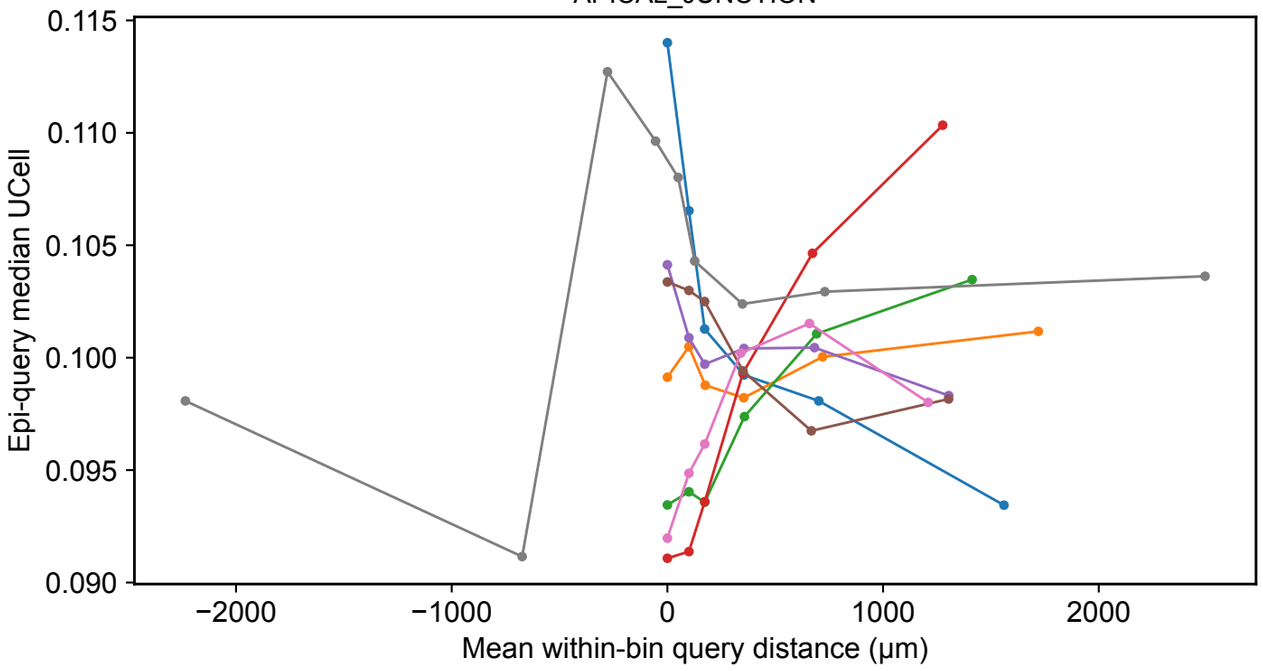
ANDROGEN_RESPONSE



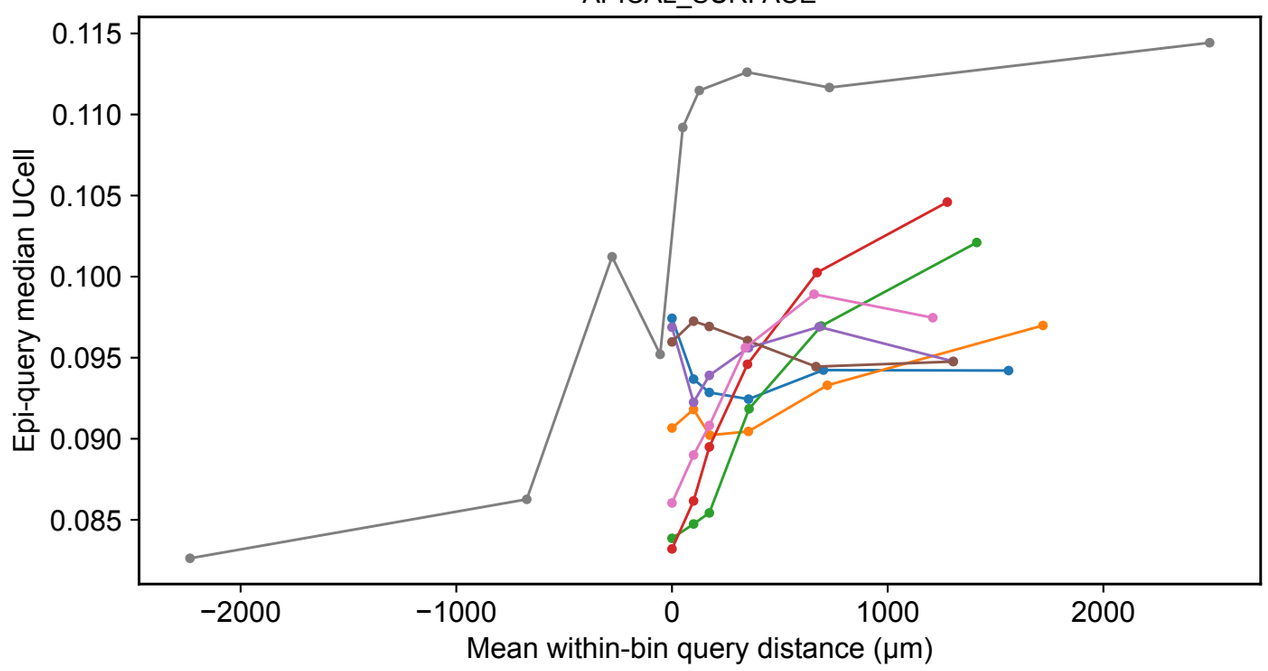
ANGIOGENESIS

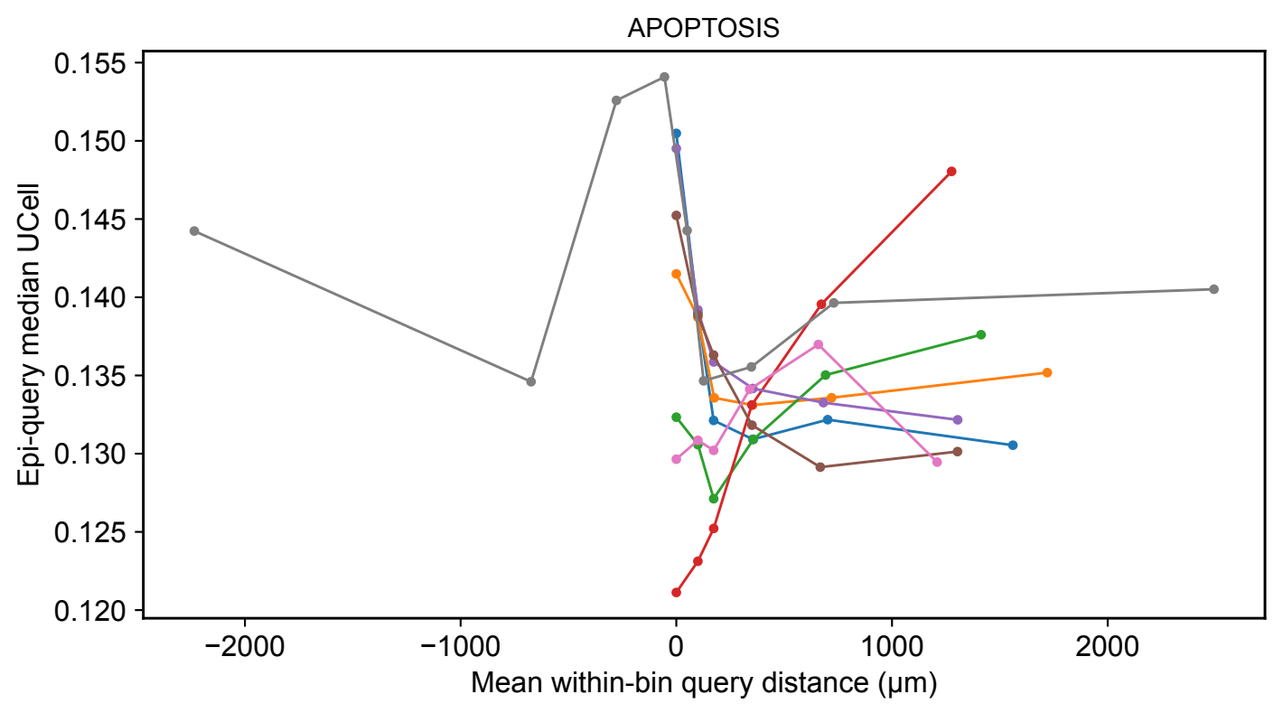


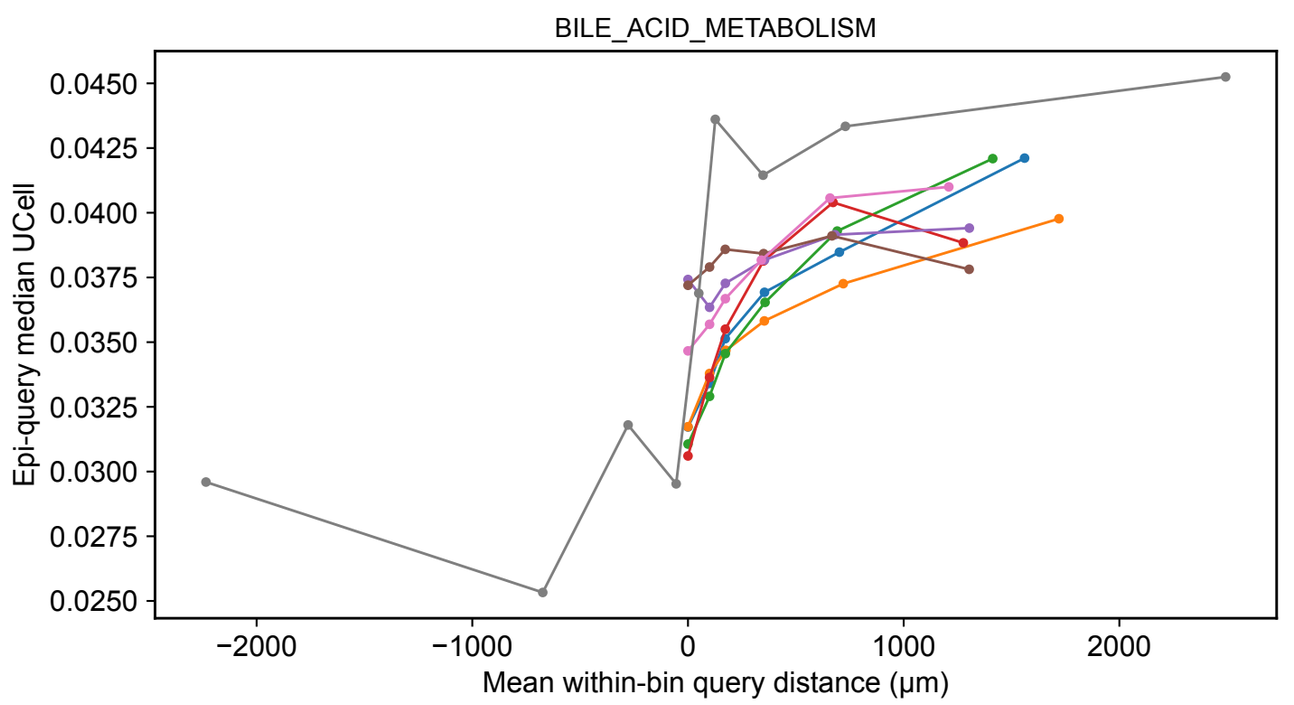
APICAL_JUNCTION

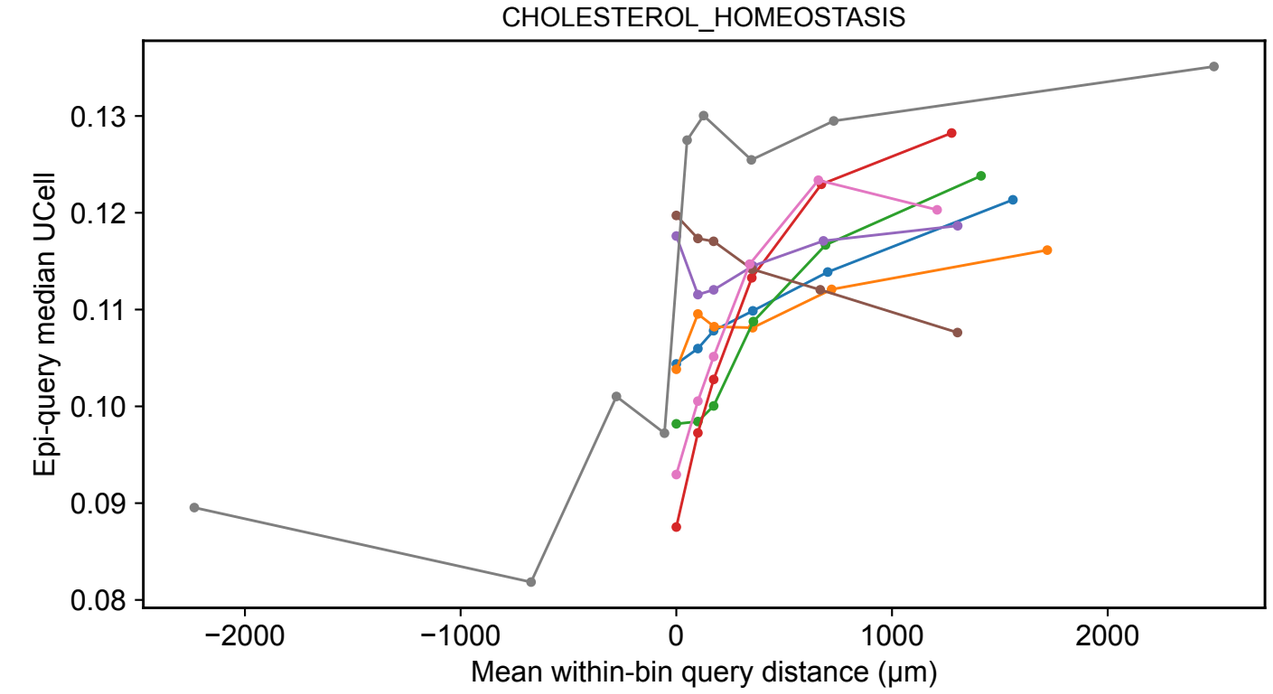


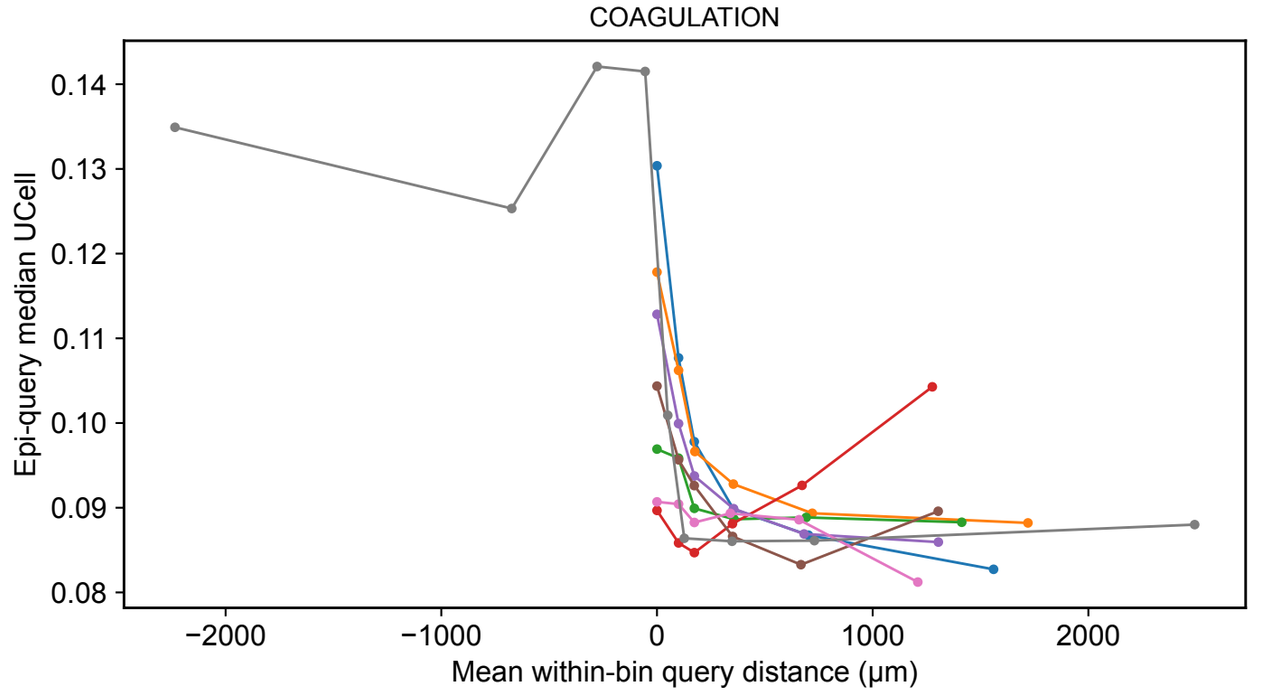
APICAL_SURFACE

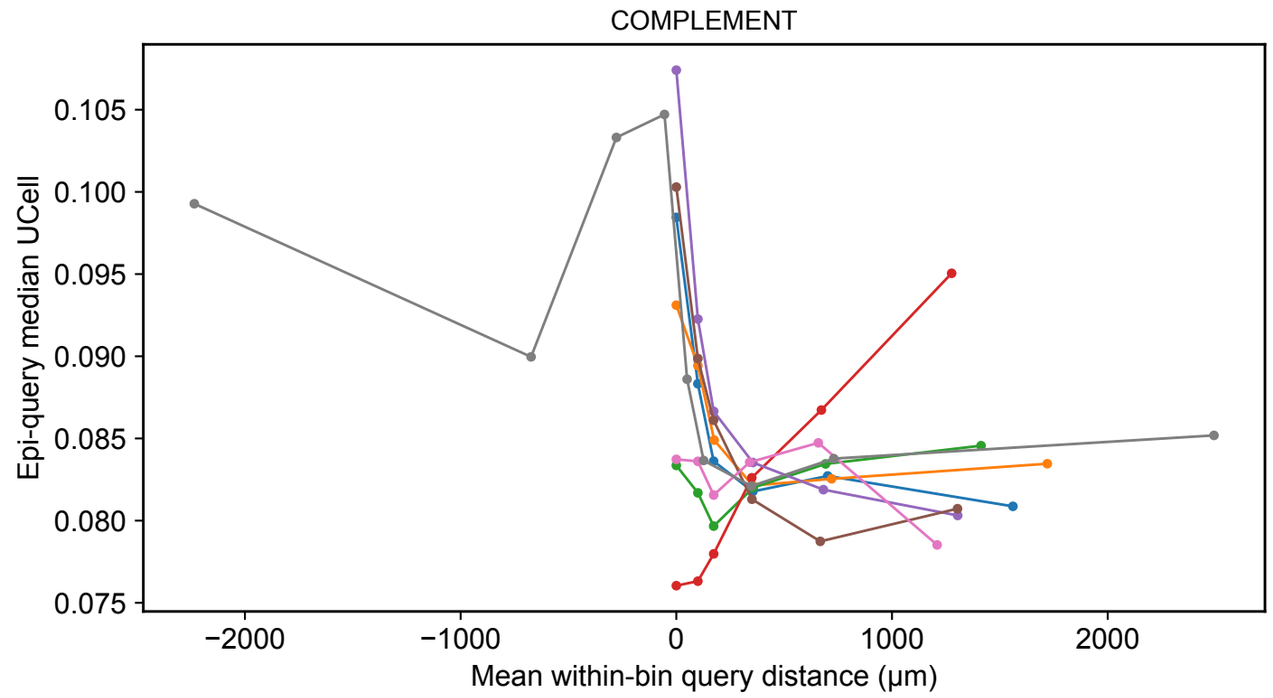


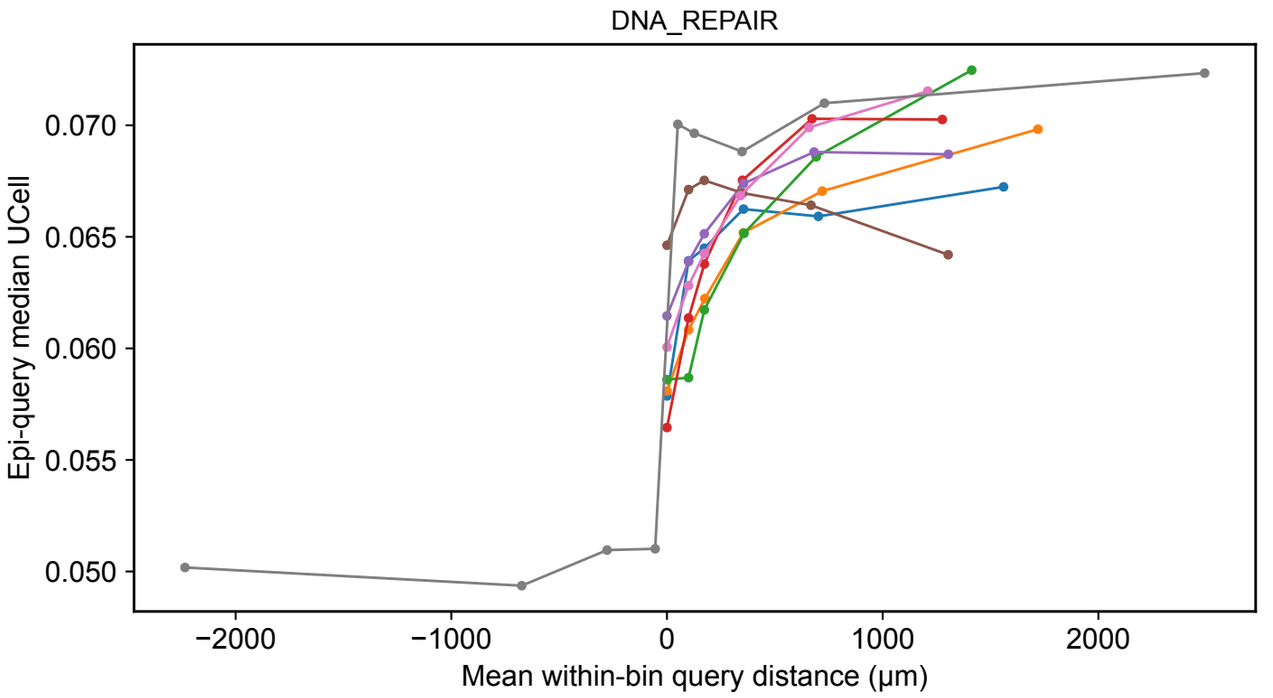

APOPTOSIS
Epi-query median UCell
0.155
0.150
0.145
0.140
0.135
0.130
0.125
0.120
−2000
−1000
0
1000
2000
Mean within-bin query distance (μm)

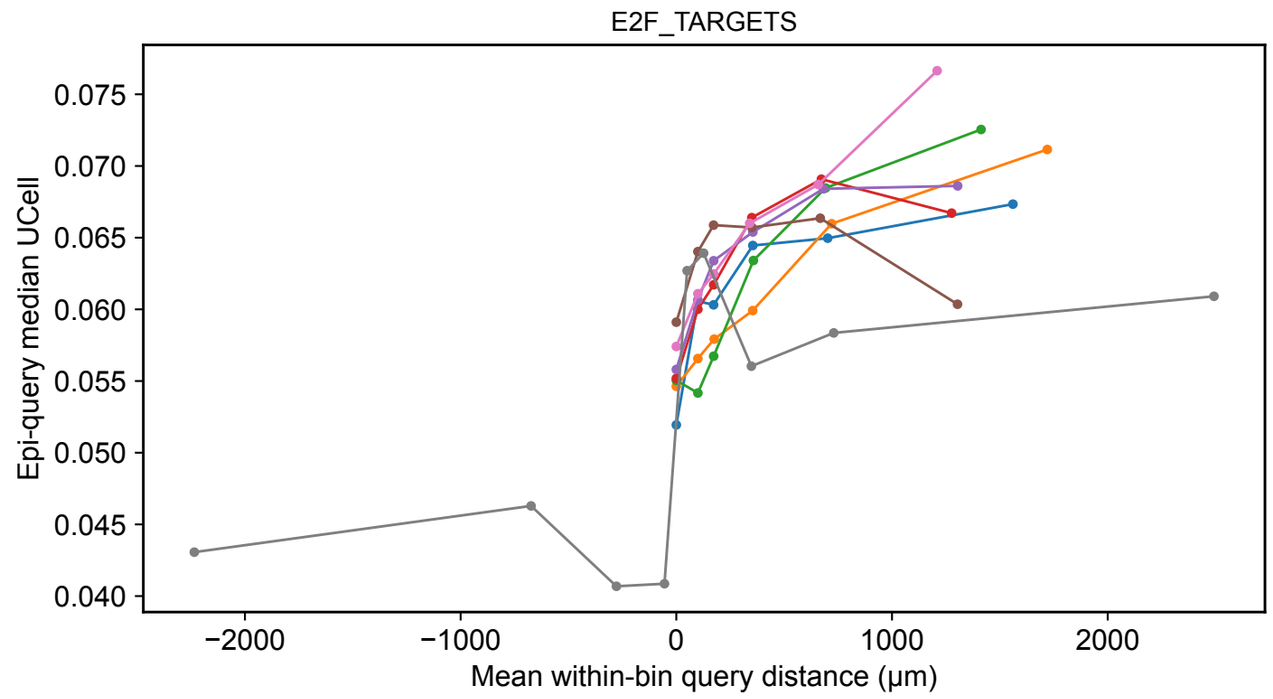

BILE_ACID_METABOLISM
Epi-query median UCell
0.0450
0.0425
0.0400
0.0375
0.0350
0.0325
0.0300
0.0275
0.0250
−2000
−1000
0
1000
2000
Mean within-bin query distance (μm)

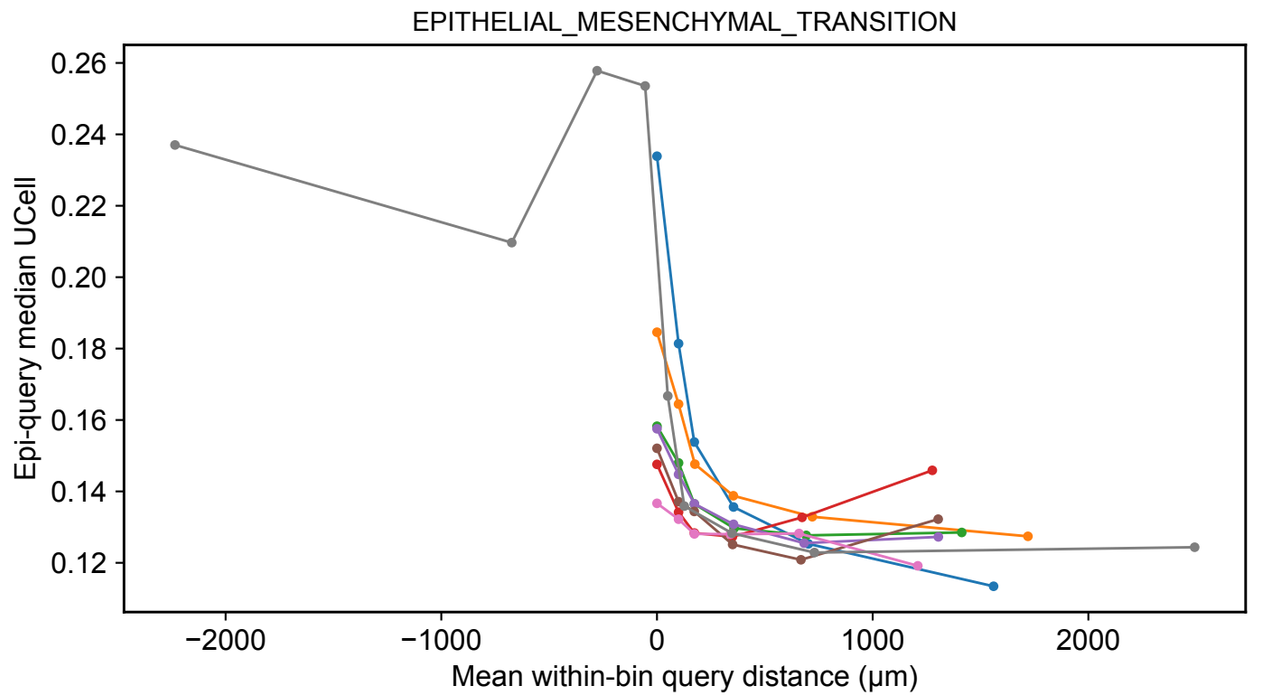

CHOLESTEROL_HOMEOSTASIS
Epi-query median UCell
0.13
0.12
0.11
0.10
0.09
0.08
−2000
−1000
0
1000
2000
Mean within-bin query distance (μm)


COAGULATION
Epi-query median UCell
0.14
0.13
0.12
0.11
0.10
0.09
0.08
−2000
−1000
0
1000
2000
Mean within-bin query distance (μm)


COMPLEMENT
Epi-query median UCell
0.105
0.100
0.095
0.090
0.085
0.080
0.075
−2000
−1000
0
1000
2000
Mean within-bin query distance (μm)

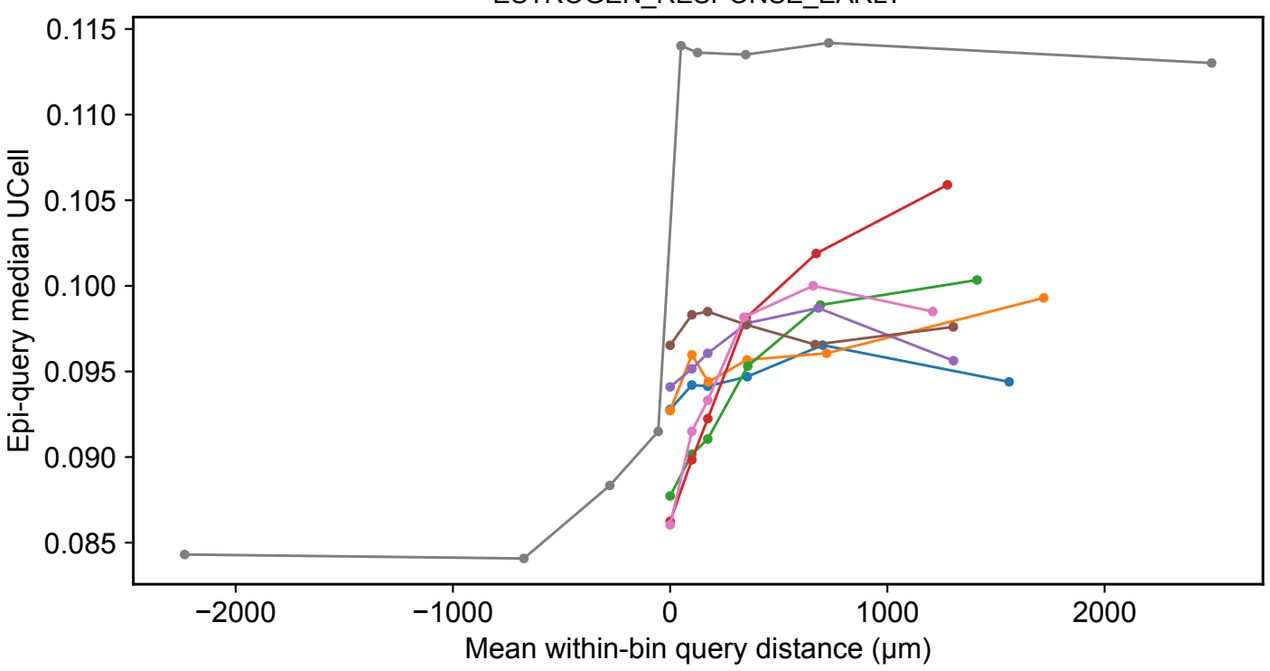

DNA_REPAIR
Epi-query median UCell
0.070
0.065
0.060
0.055
0.050
−2000
−1000
0
1000
2000
Mean within-bin query distance (μm)


E2F_TARGETS
Epi-query median UCell
0.075
0.070
0.065
0.060
0.055
0.050
0.045
0.040
−2000
−1000
0
1000
2000
Mean within-bin query distance (μm)

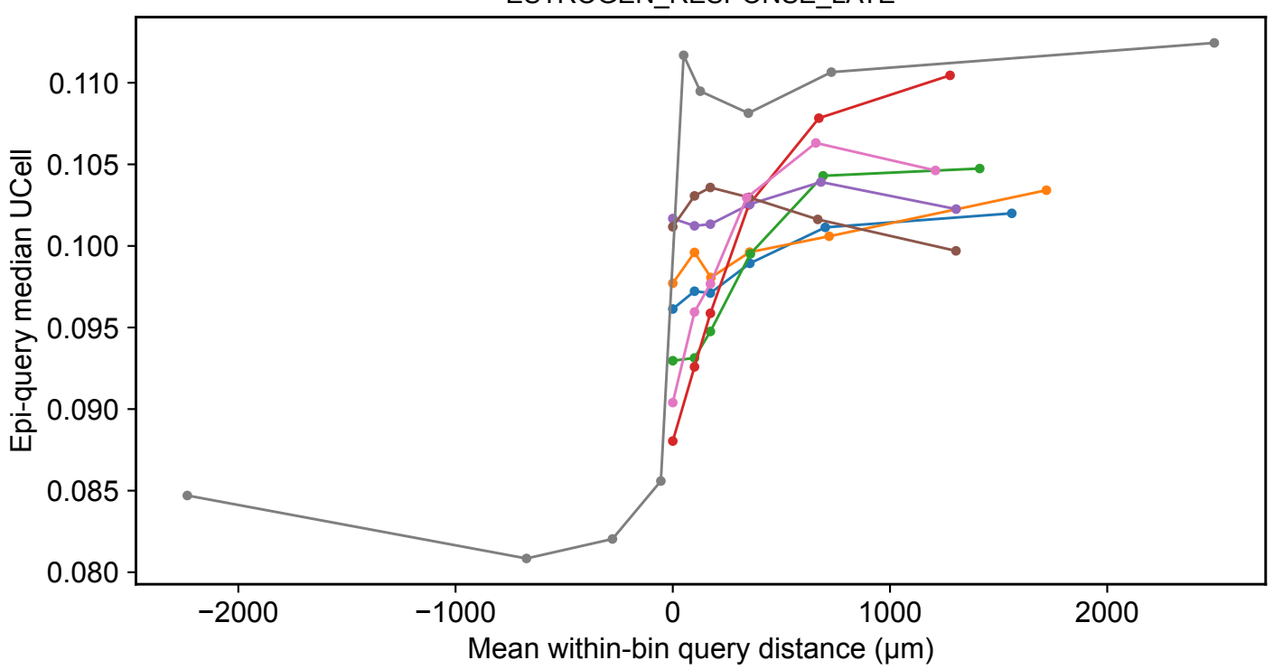

EPITHELIAL_MESENCHYMAL_TRANSITION
Epi-query median UCell
0.26
0.24
0.22
0.20
0.18
0.16
0.14
0.12
−2000
−1000
0
1000
2000
Mean within-bin query distance (μm)

Matrix_CAF_P95
iCAF_P95
rCAF_P95
vCAF_P95
C1QC TAM_P95
SPP1 TAM_P95
Treg_P95
local_interface

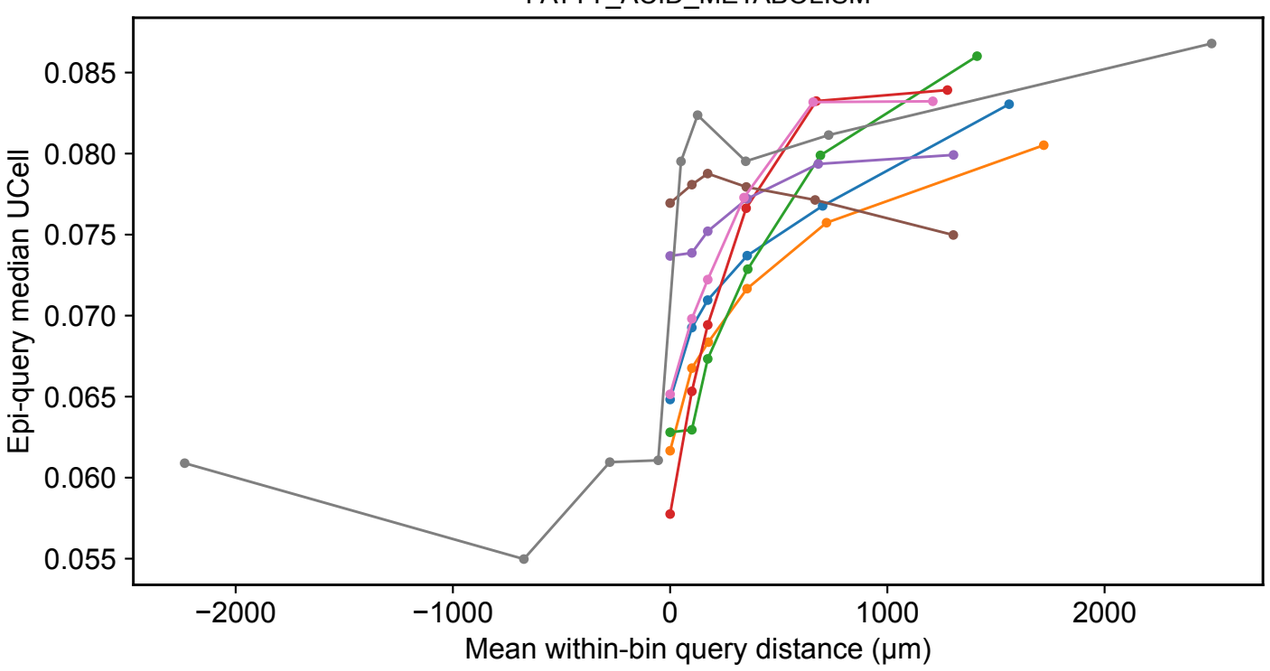
ESTROGEN_RESPONSE_EARLY



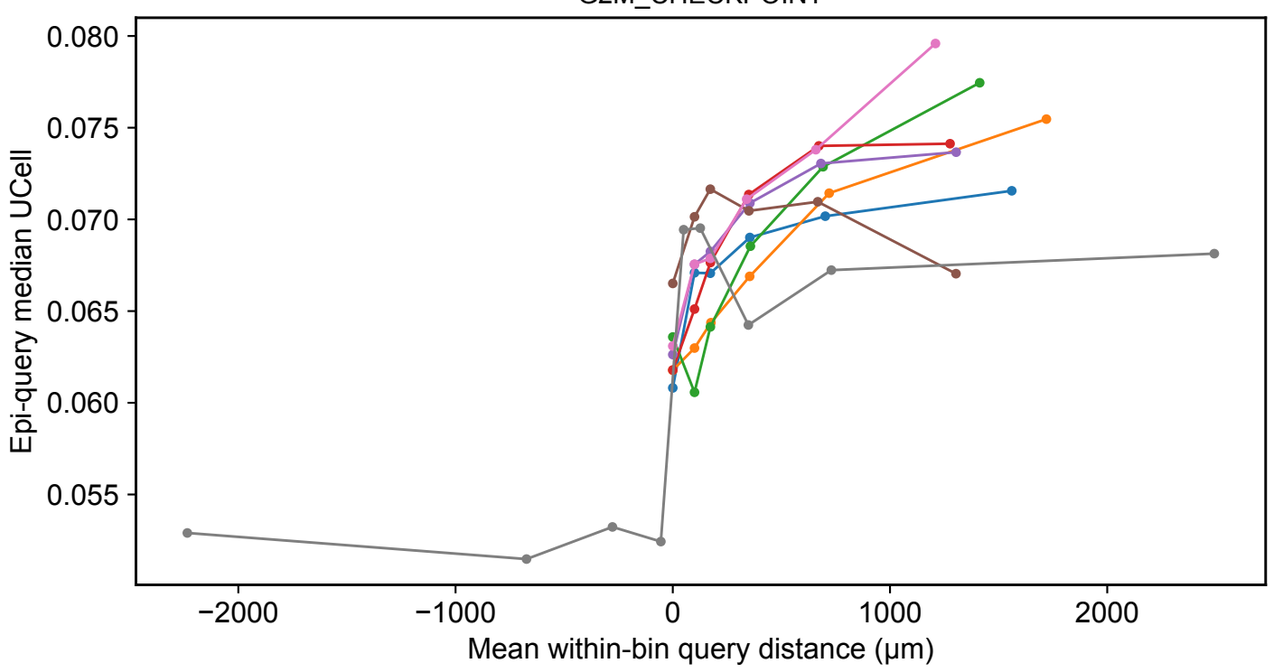
ESTROGEN_RESPONSE_LATE



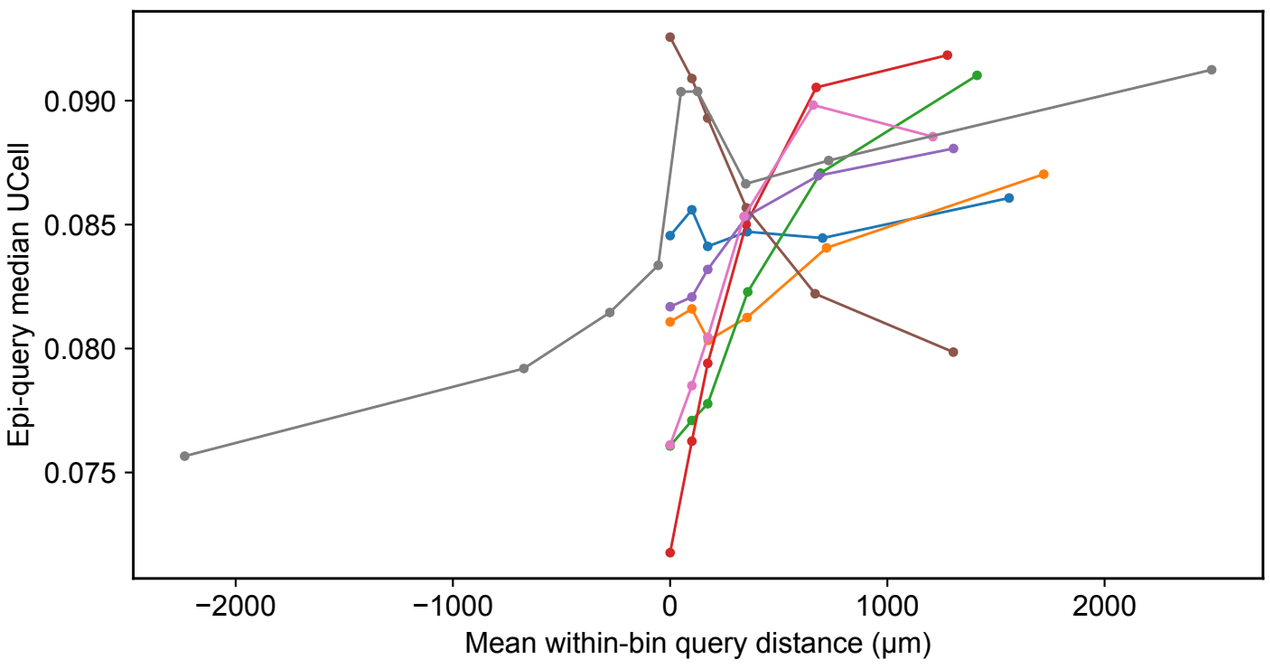
FATTY_ACID_METABOLISM



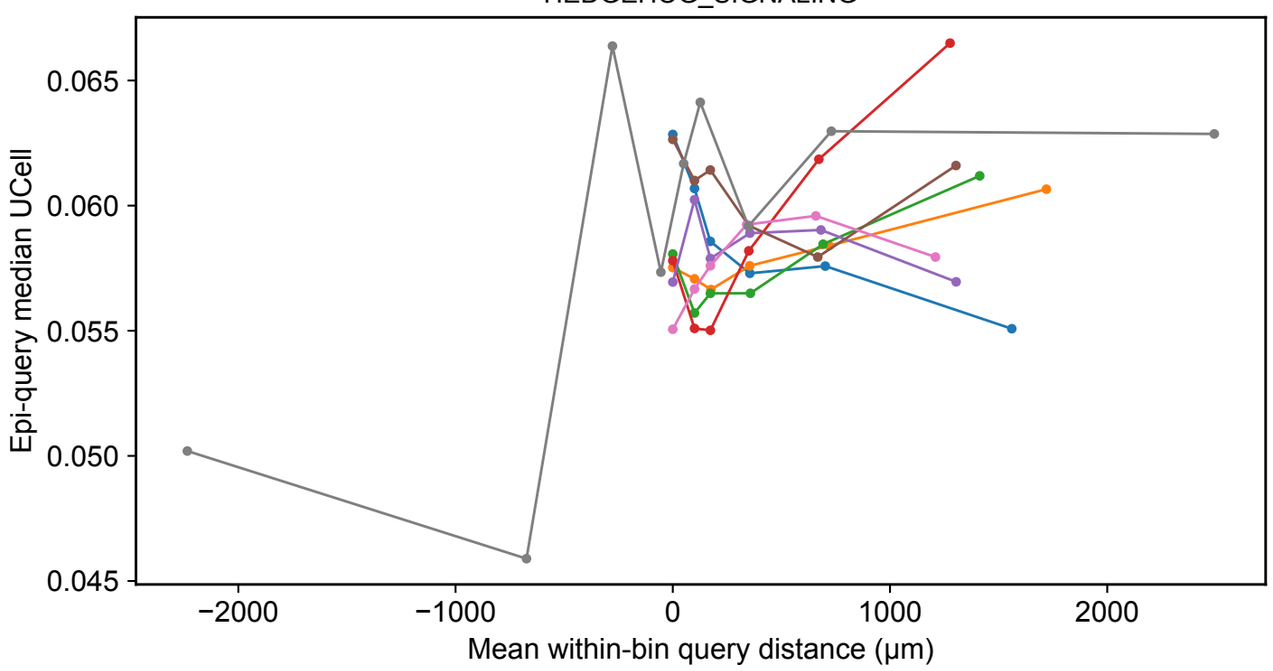
G2M_CHECKPOINT



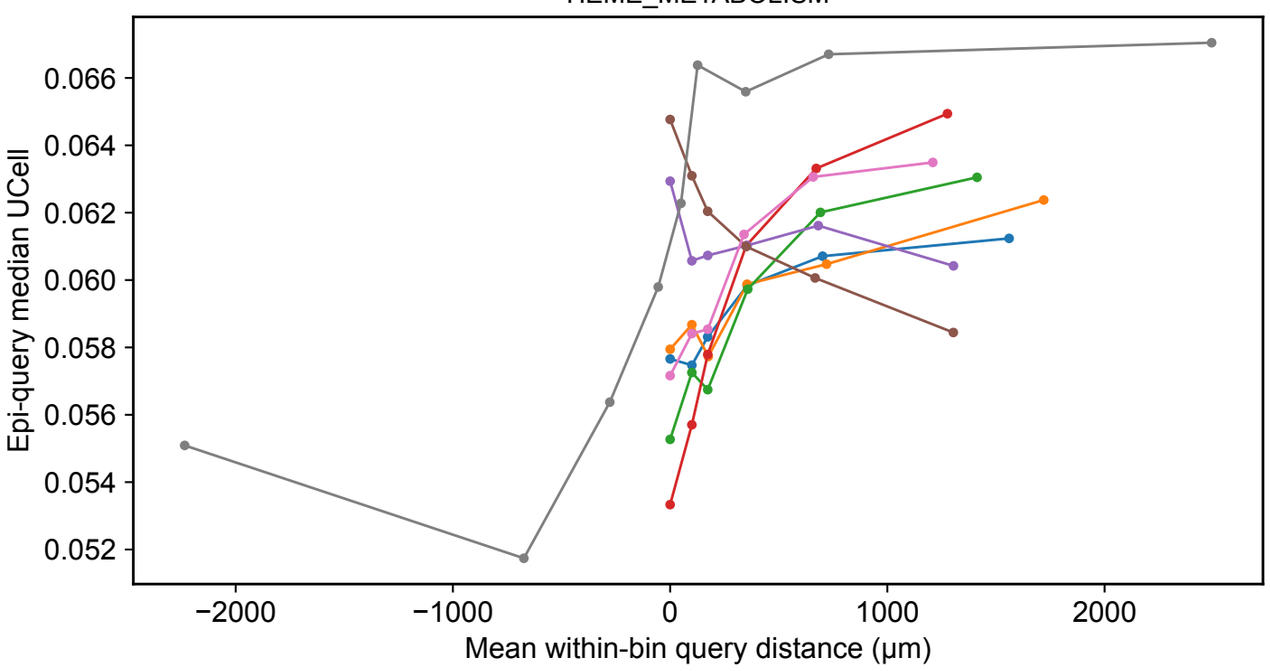
GLYCOLYSIS



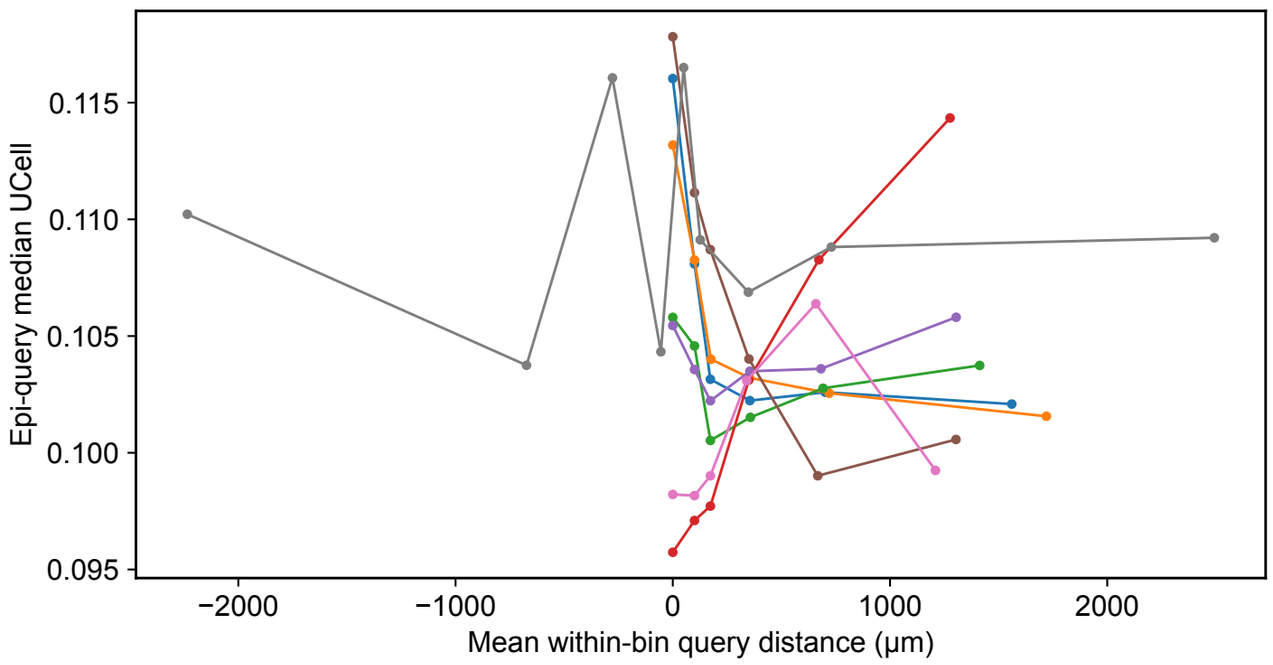
HEDGEHOG_SIGNALING



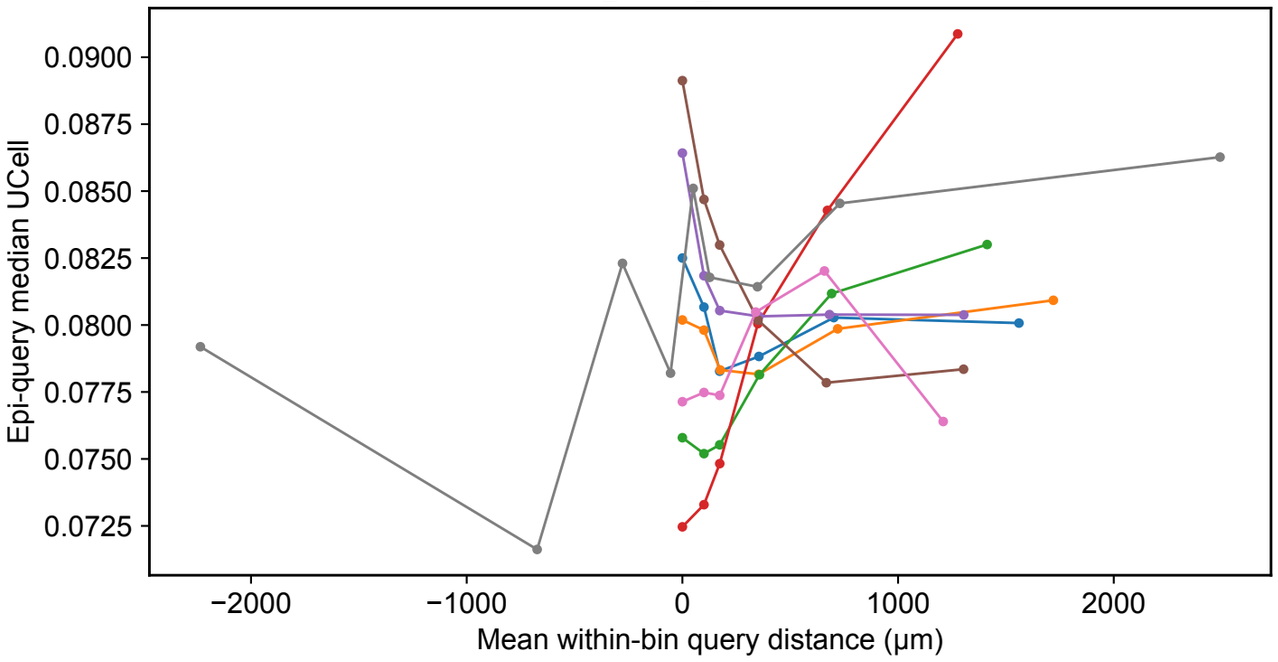
HEME_METABOLISM



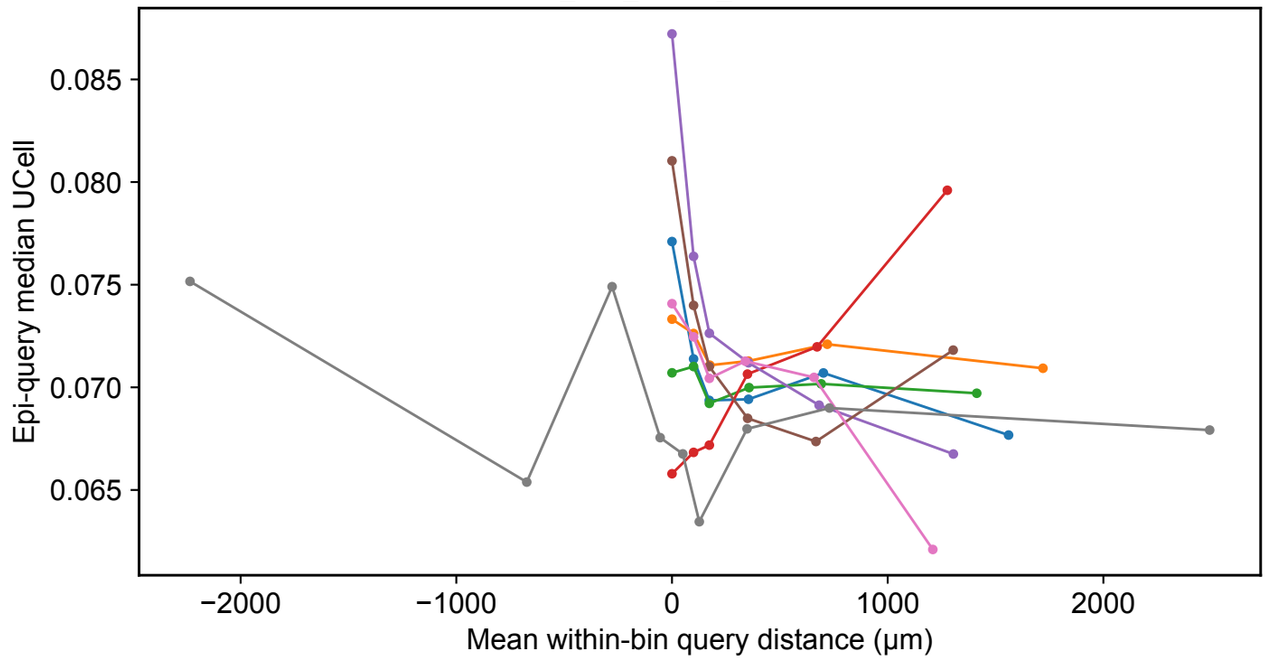
HYPOXIA



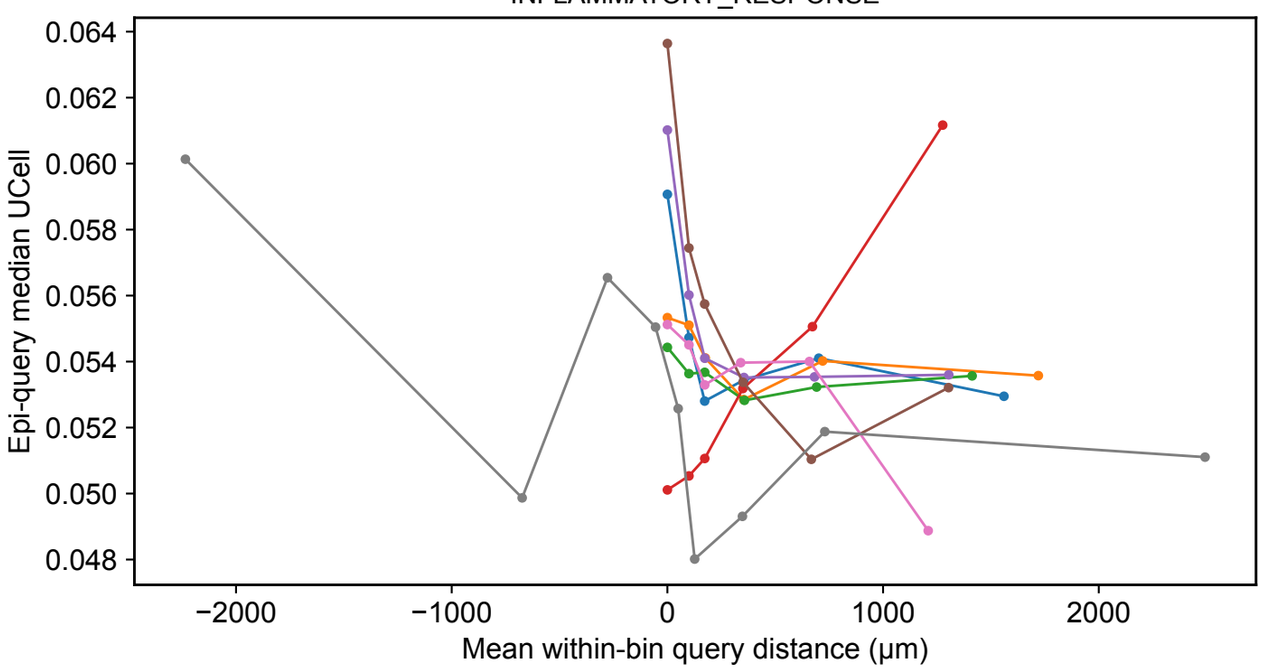
IL2_STAT5_SIGNALING



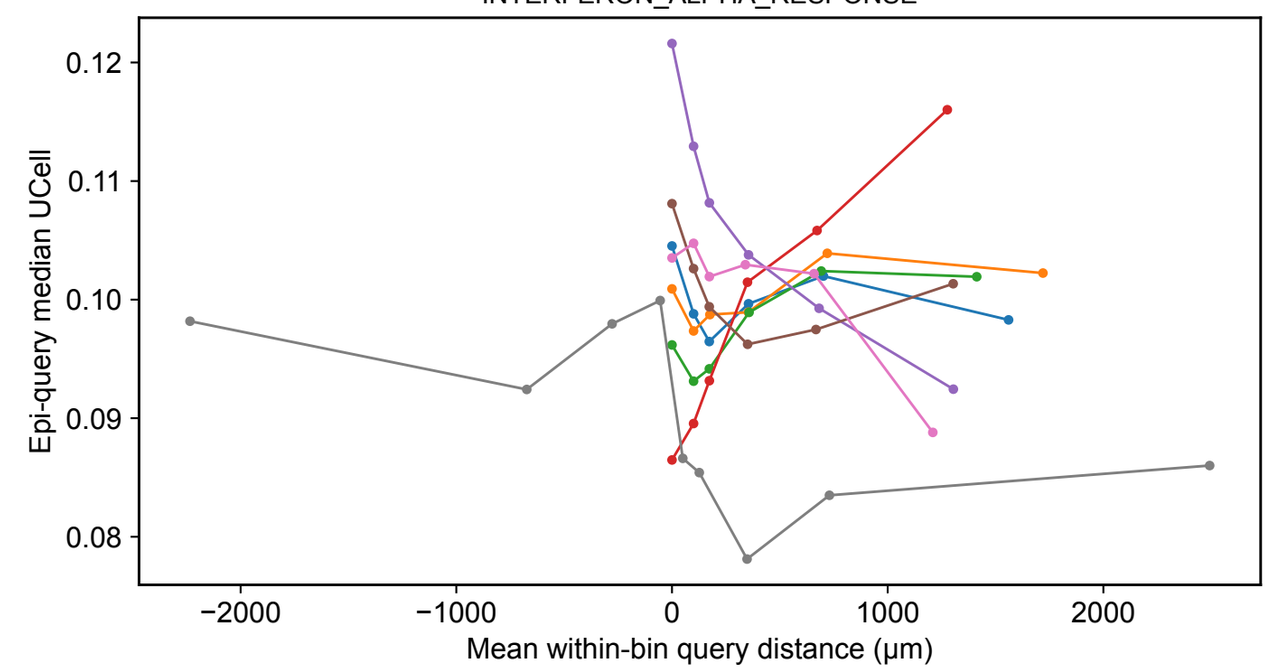
IL6_JAK_STAT3_SIGNALING



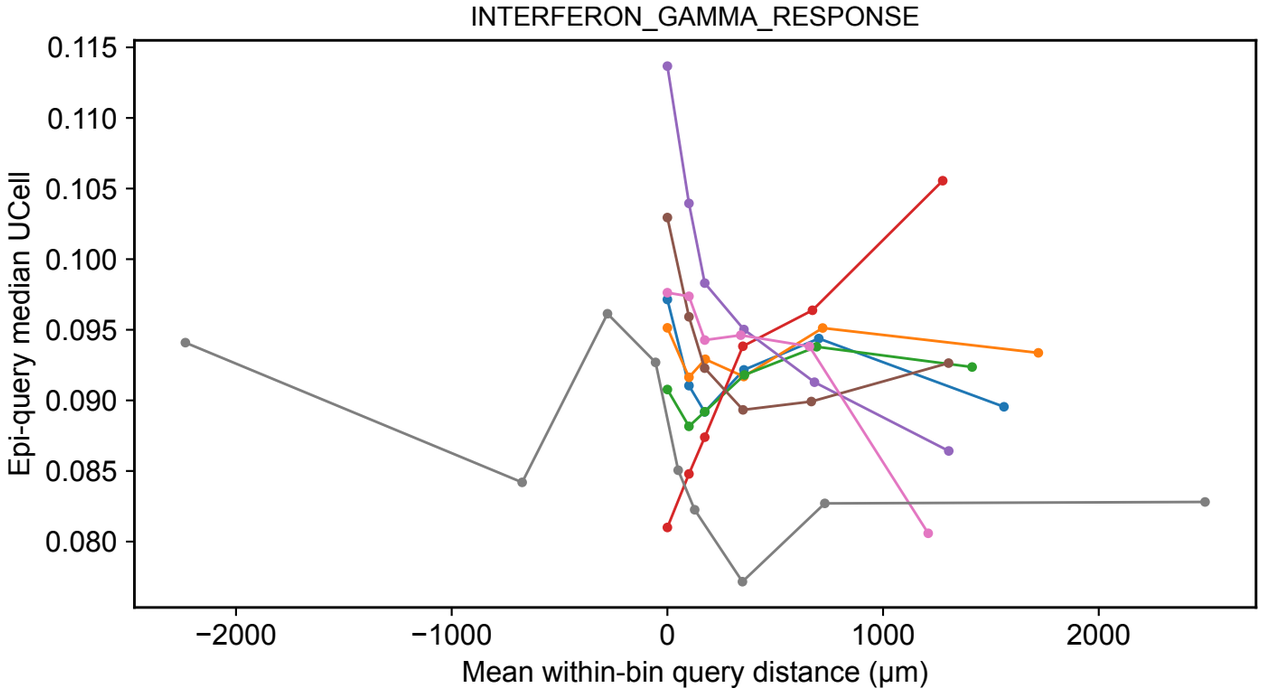
INFLAMMATORY_RESPONSE



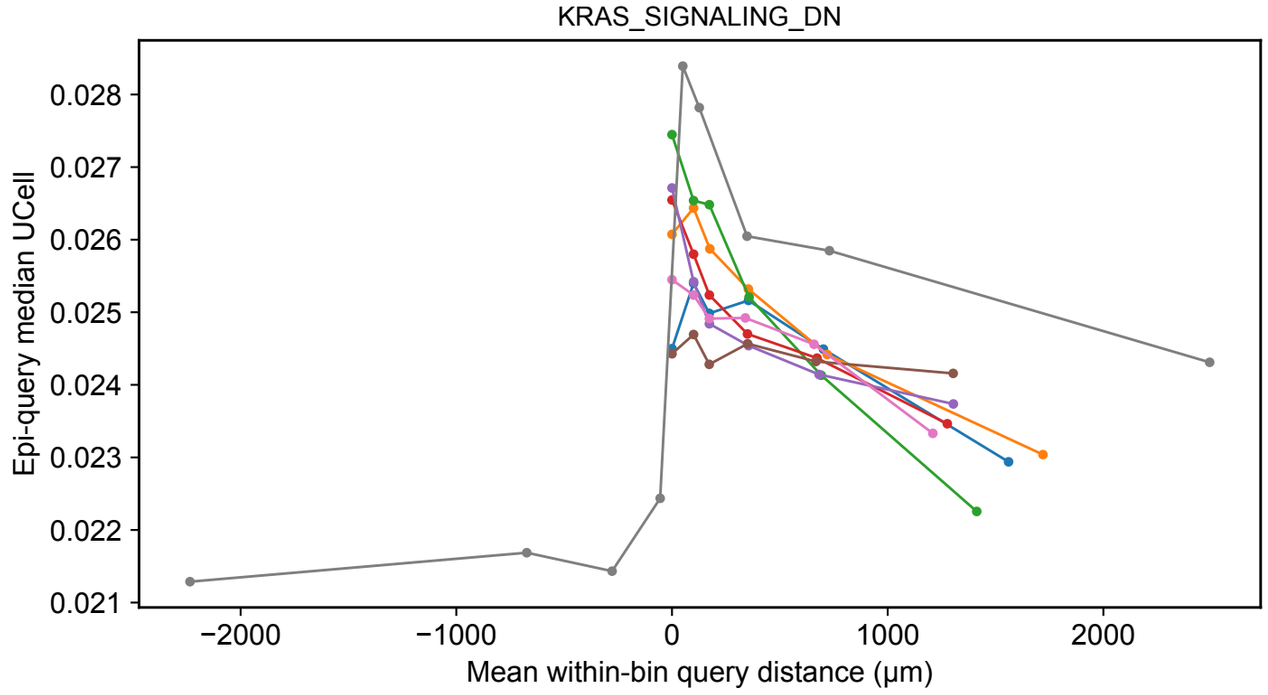
INTERFERON_ALPHA_RESPONSE



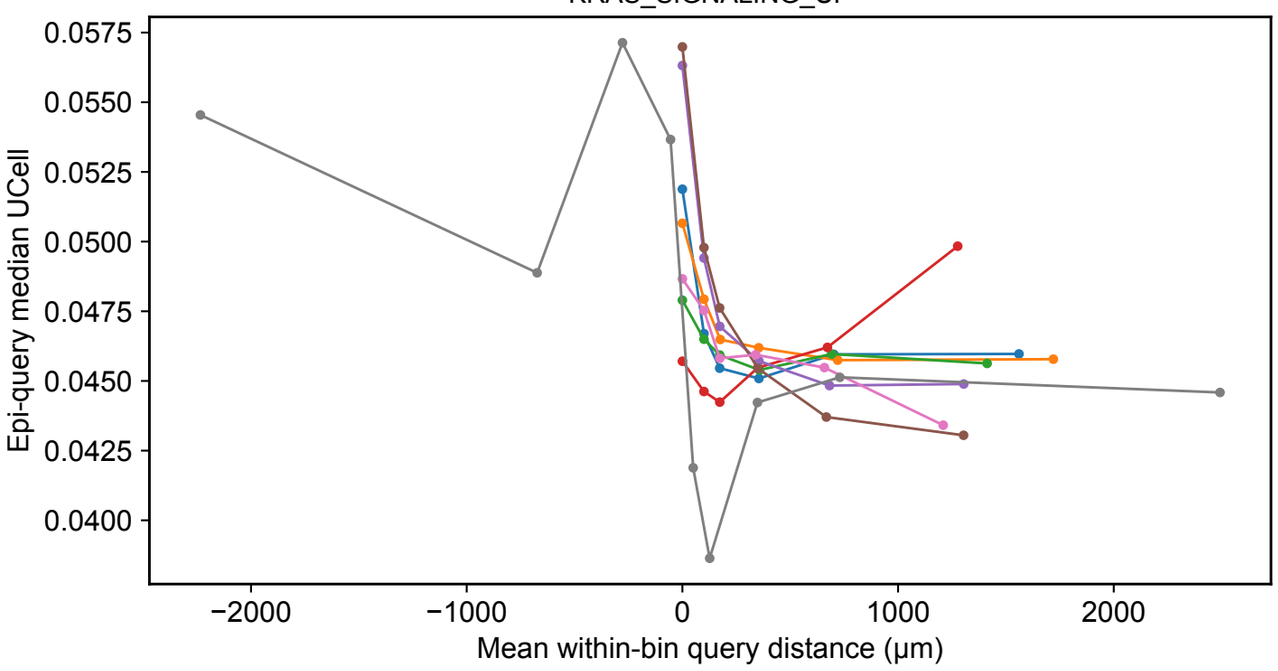
INTERFERON_GAMMA_RESPONSE



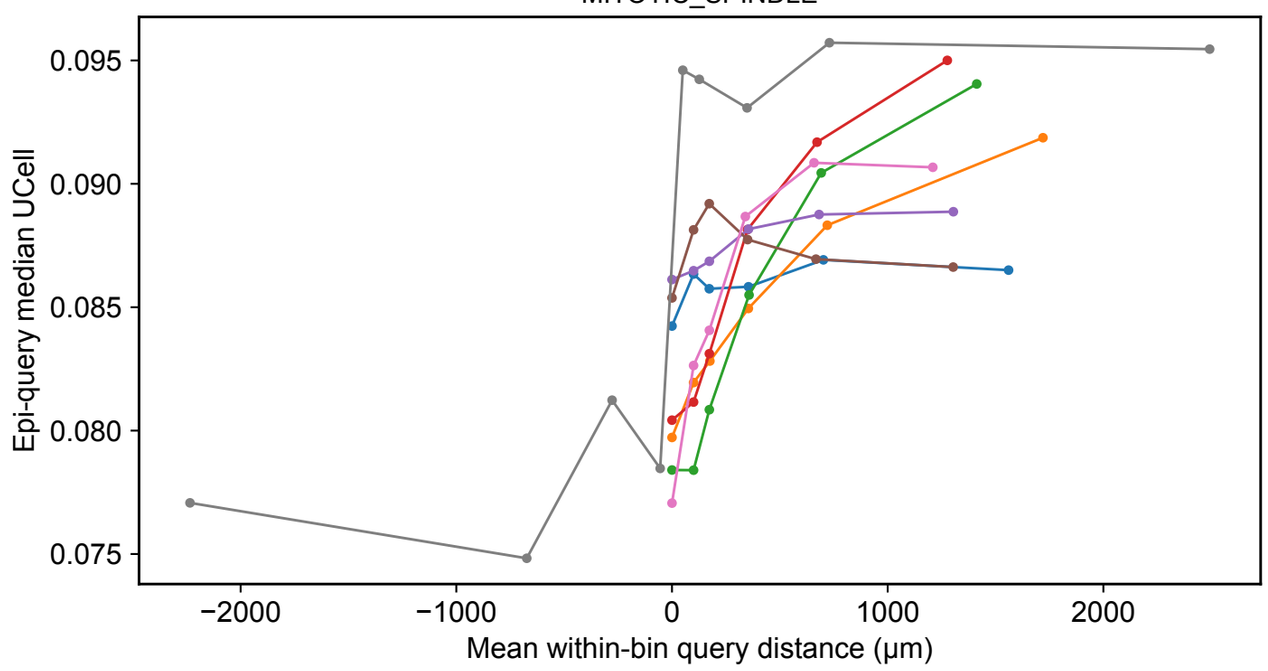
KRAS_SIGNALING_DN



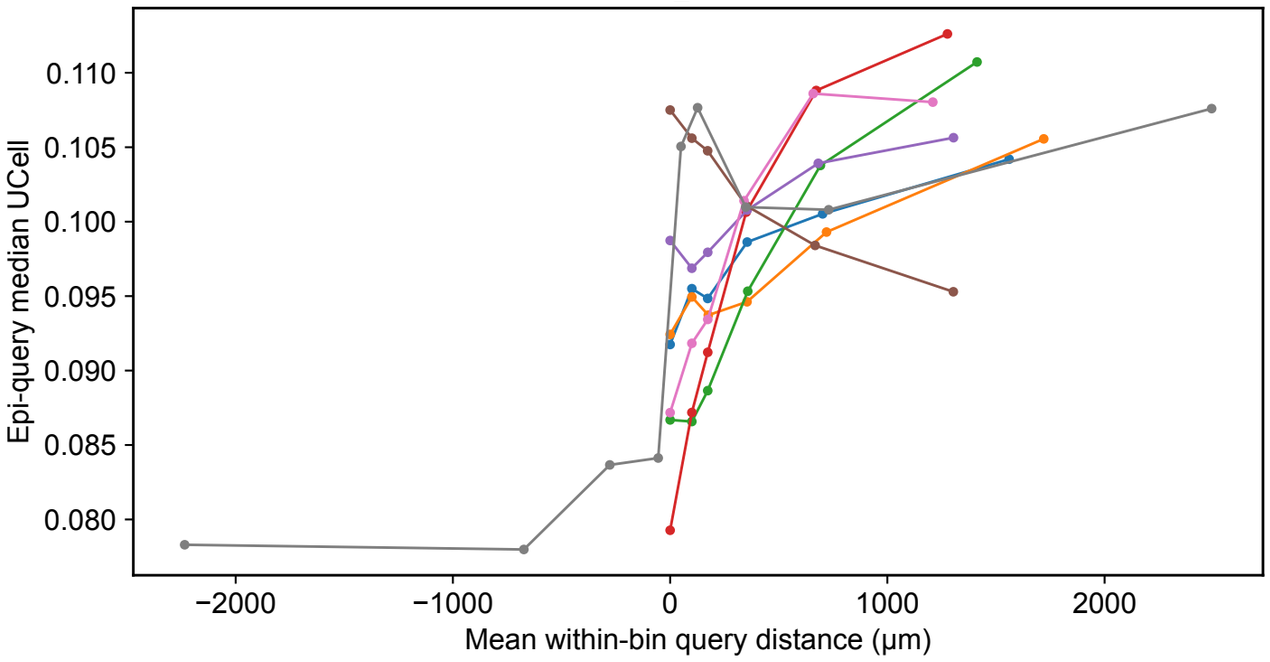
KRAS_SIGNALING_UP



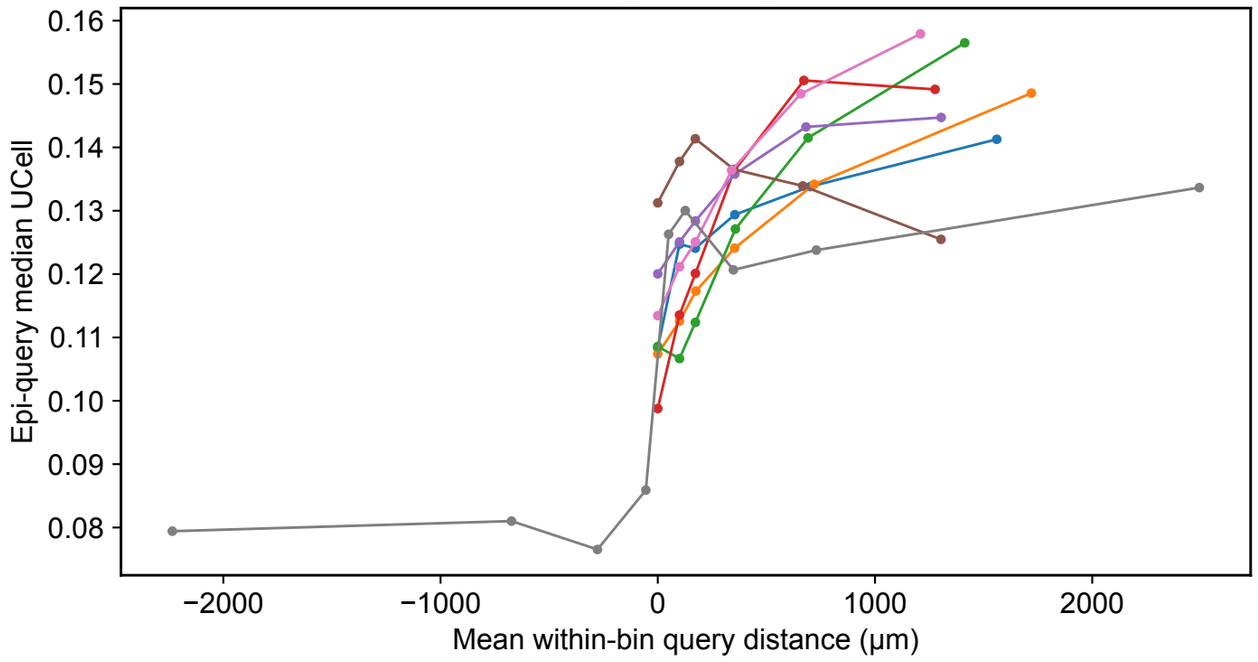
MITOTIC_SPINDLE



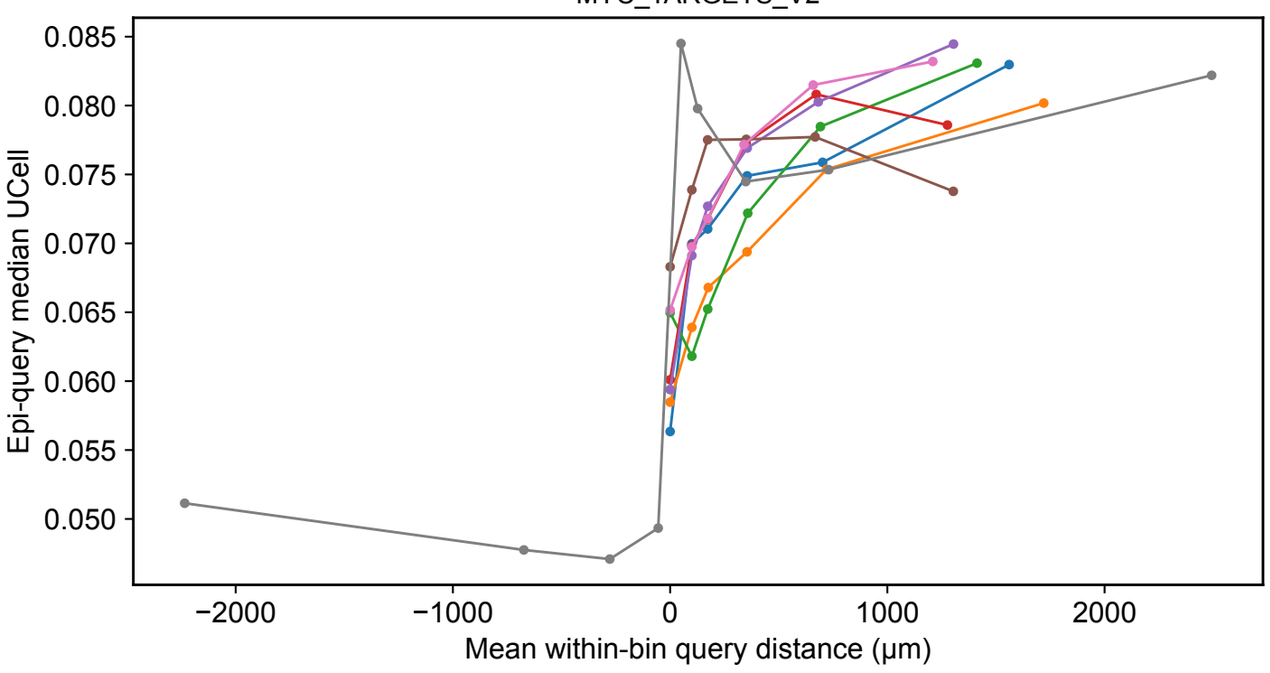
MTORC1_SIGNALING



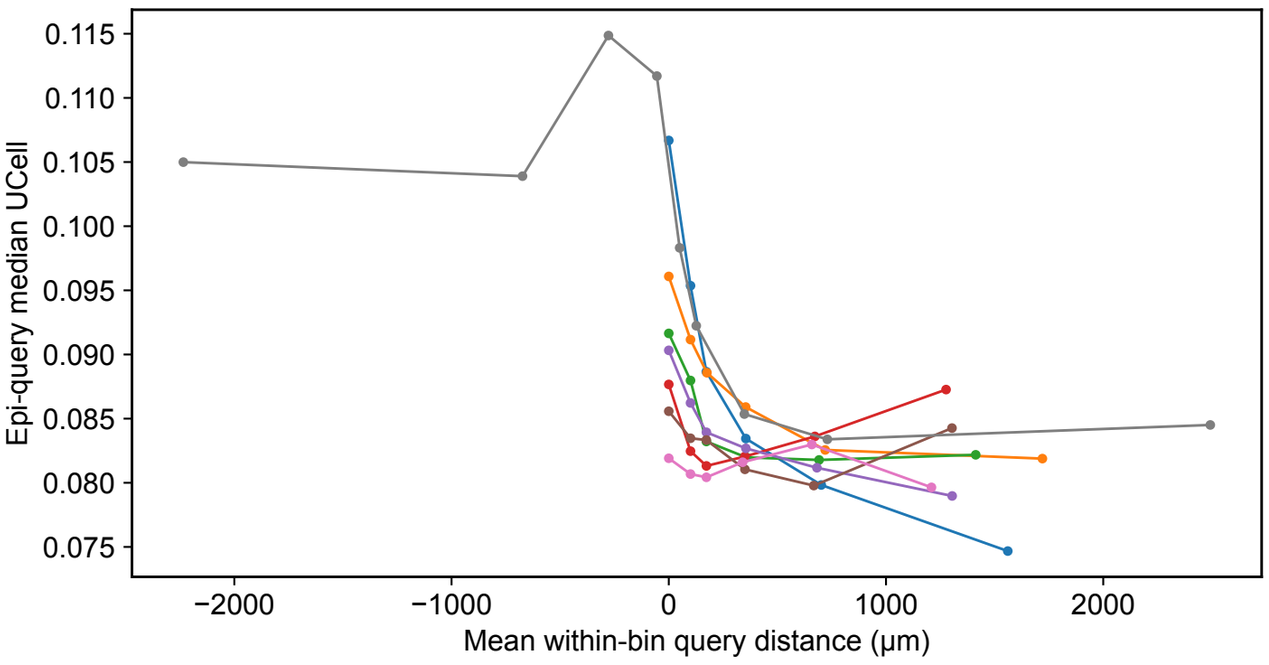
MYC_TARGETS_V1



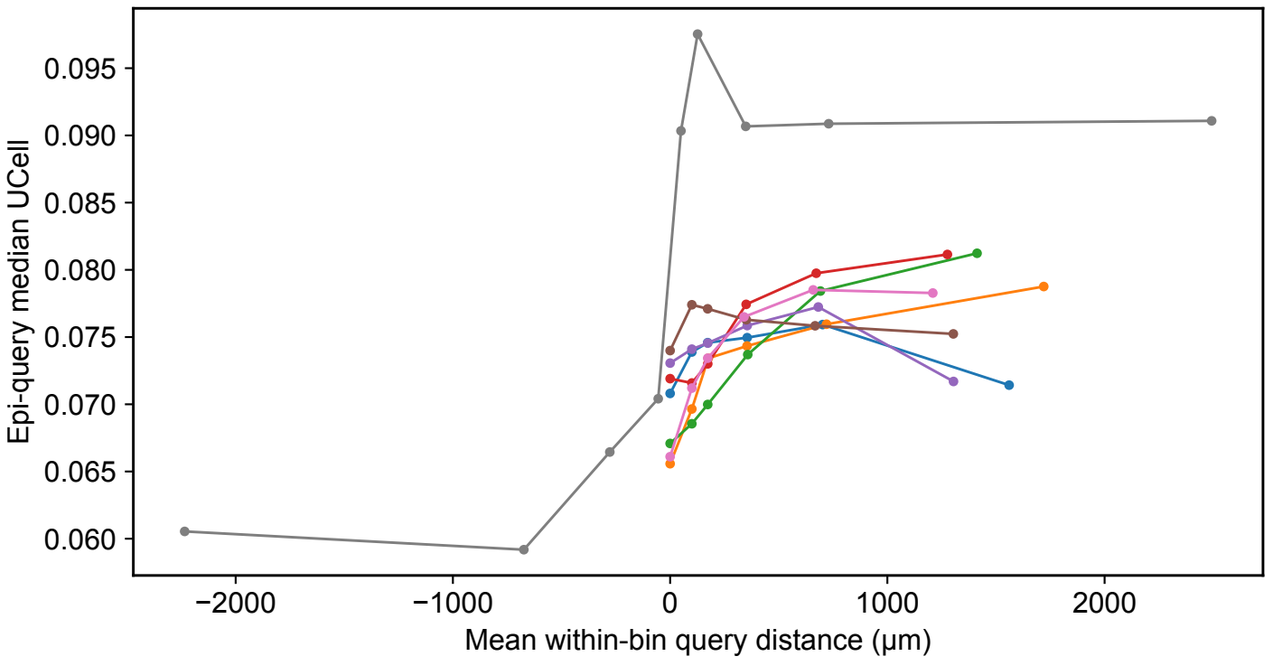
MYC_TARGETS_V2



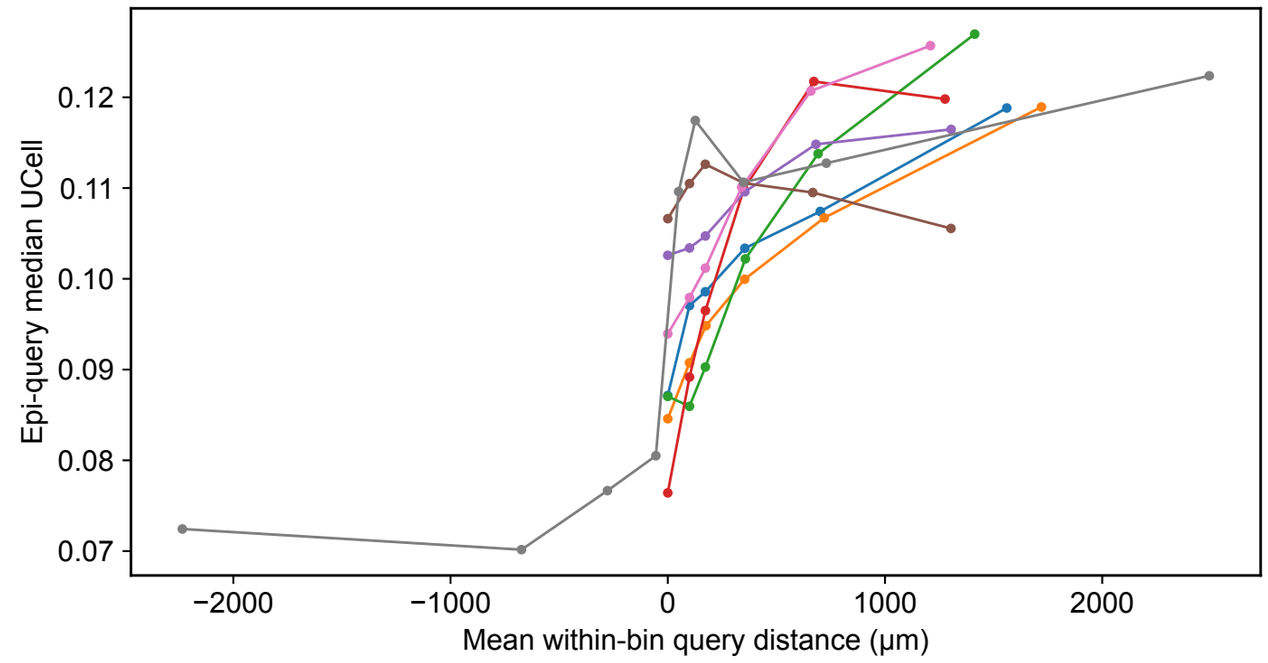
MYOGENESIS



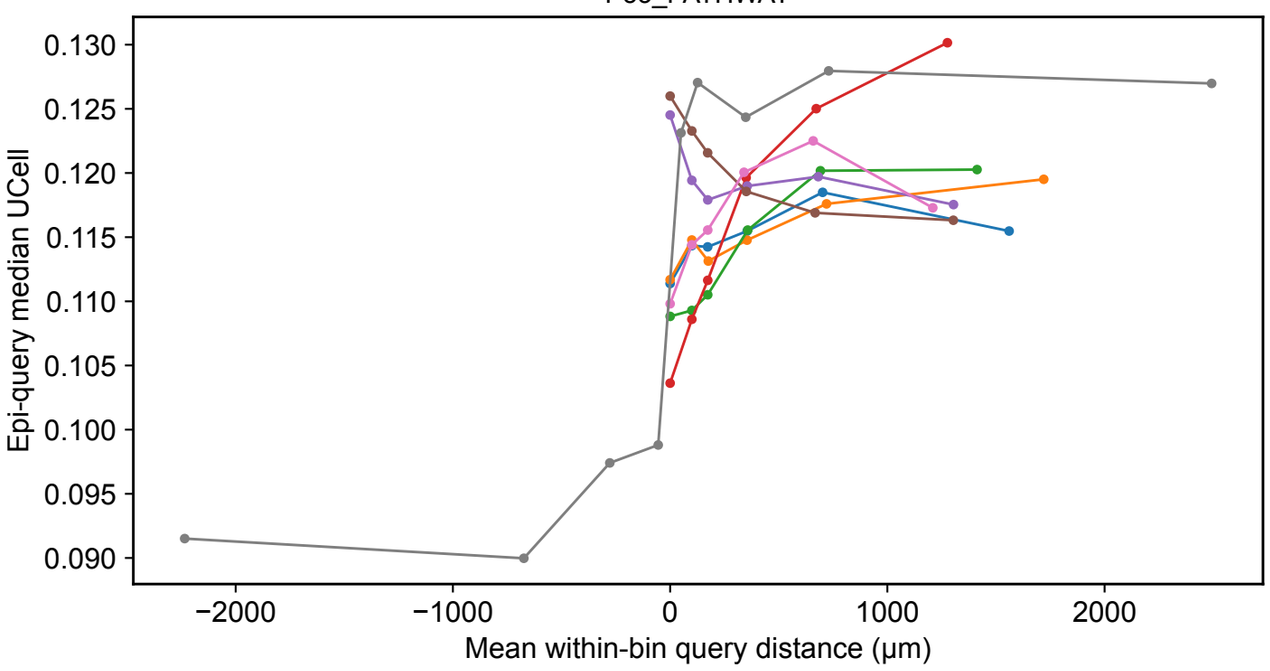
NOTCH_SIGNALING



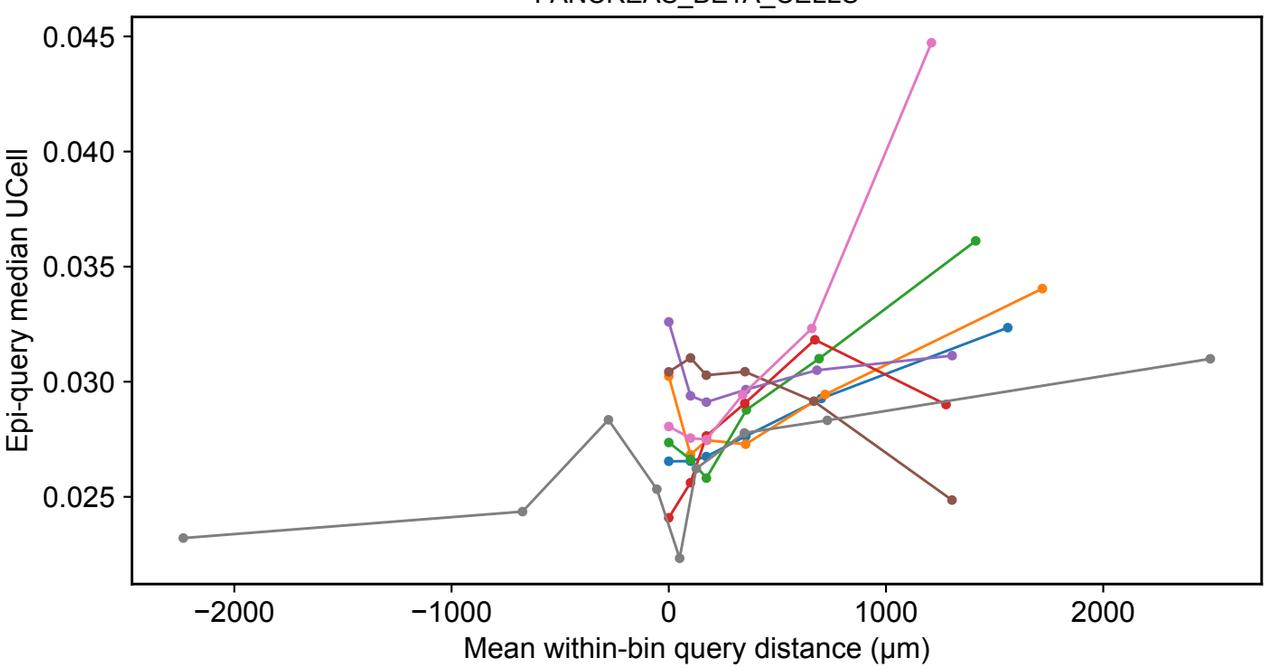
OXIDATIVE_PHOSPHORYLATION



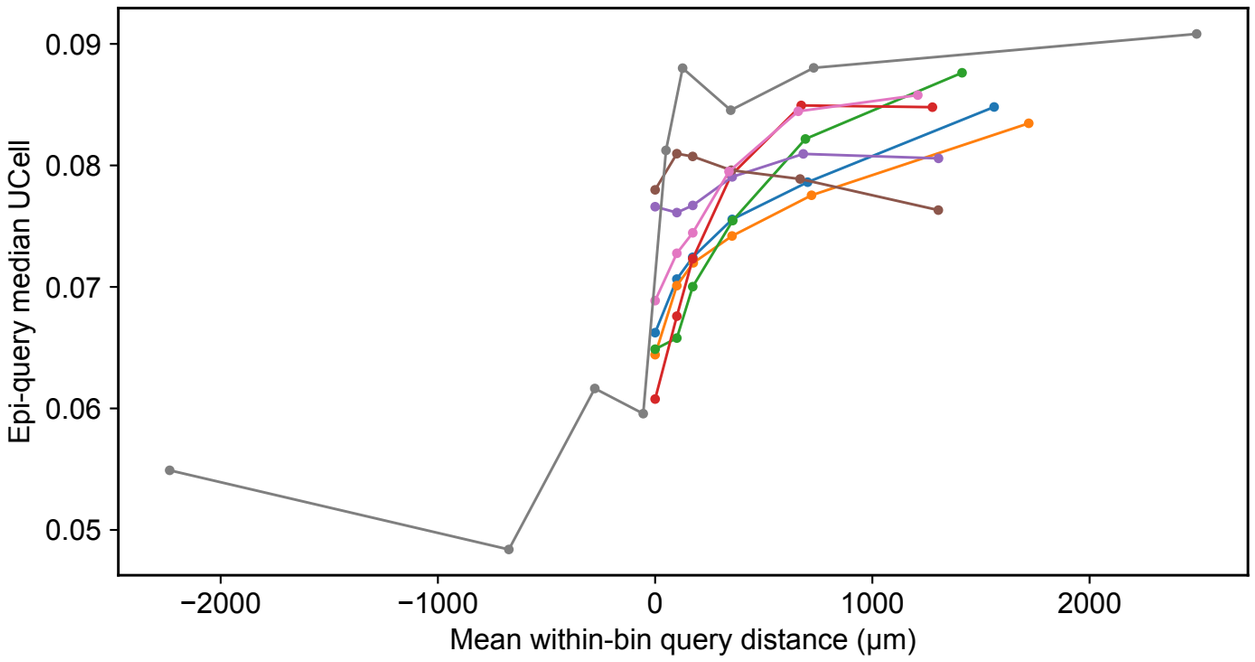
P53_PATHWAY



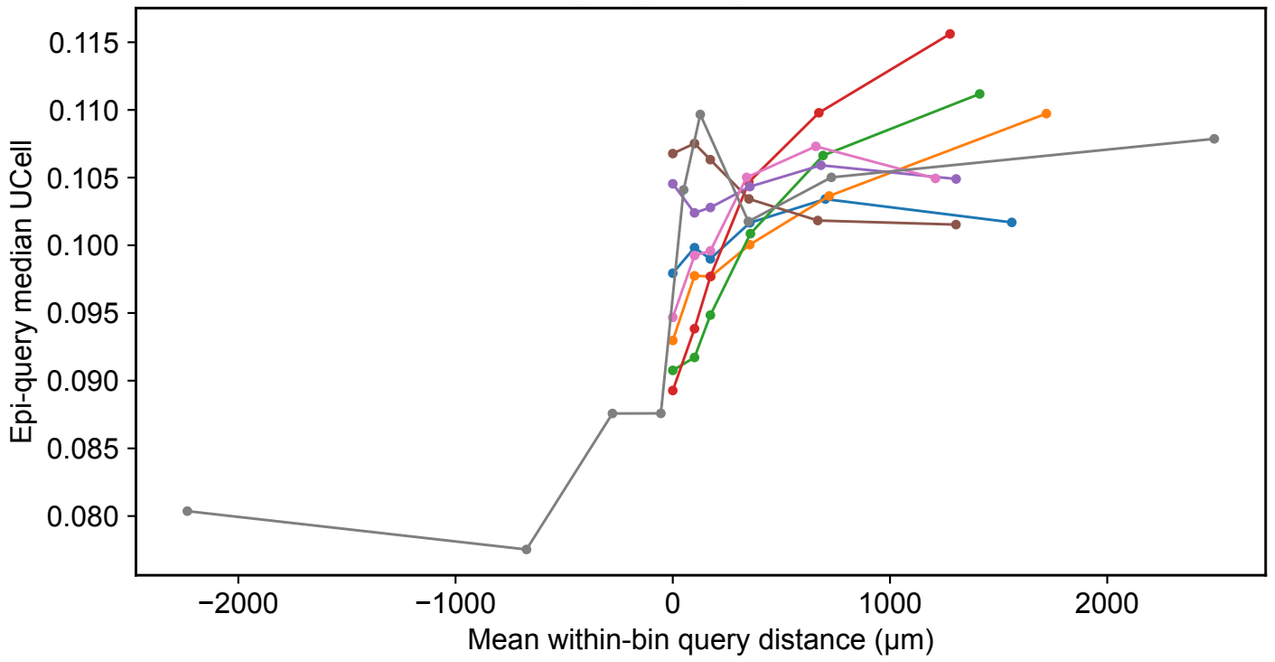
PANCREAS_BETA_CELLS



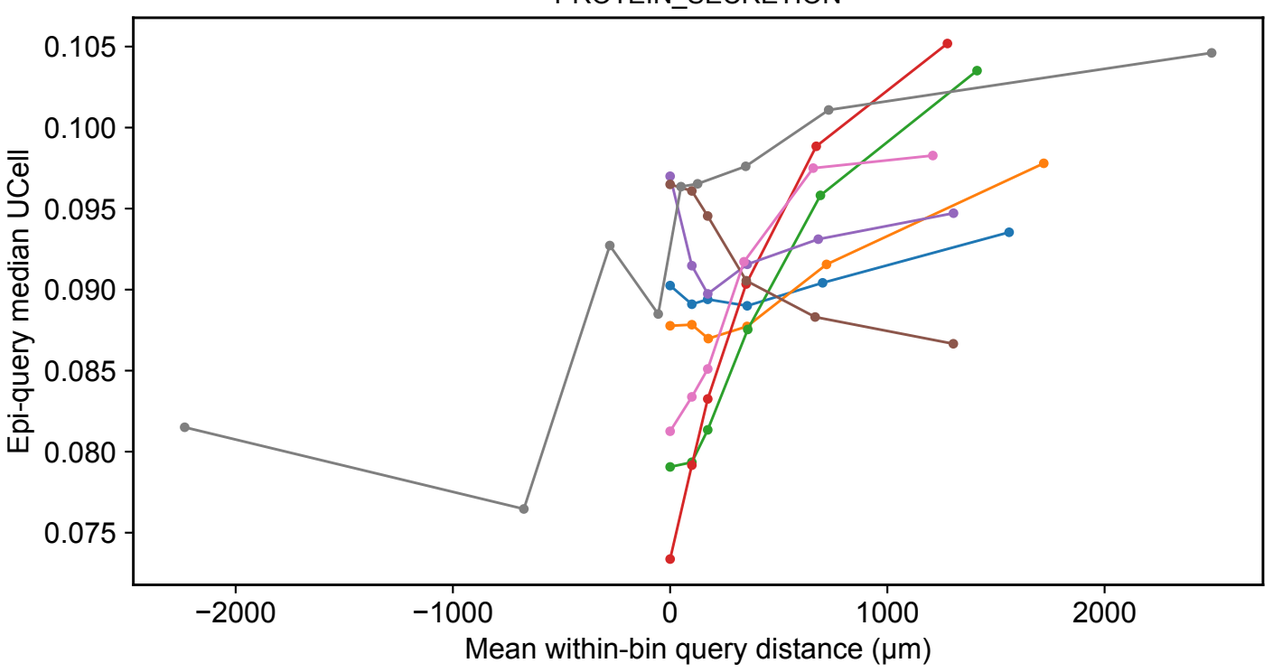
PEROXISOME



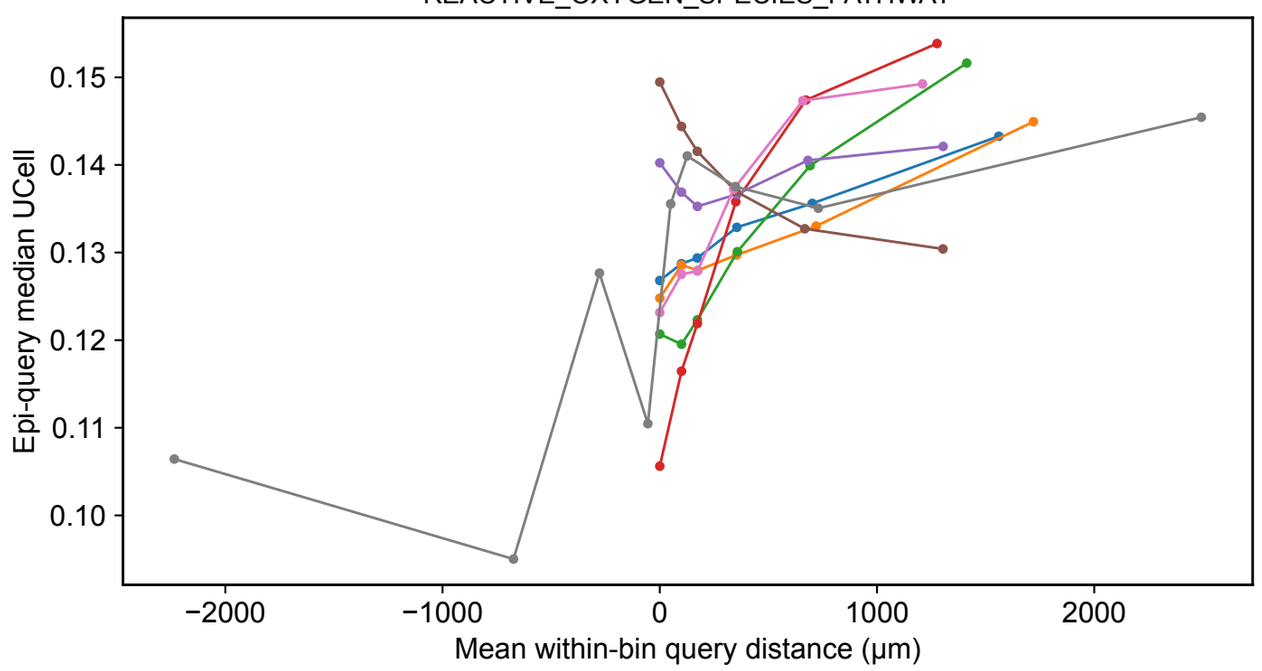
PI3K_AKT_MTOR_SIGNALING



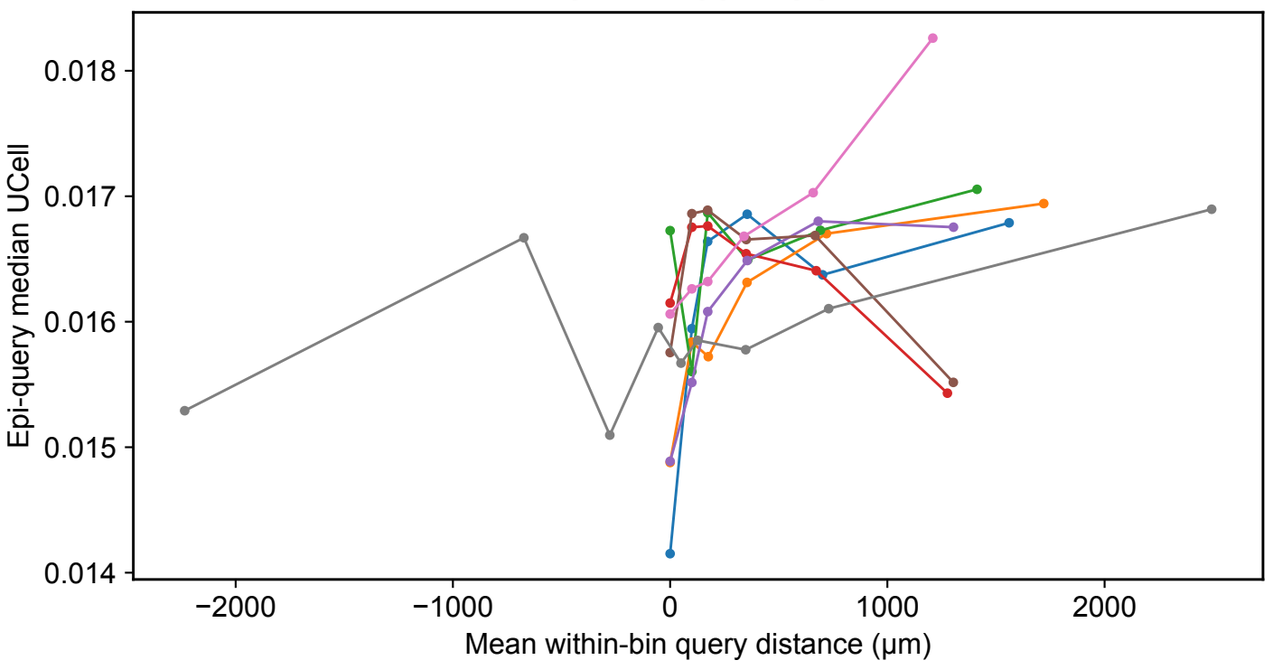
PROTEIN_SECRETION



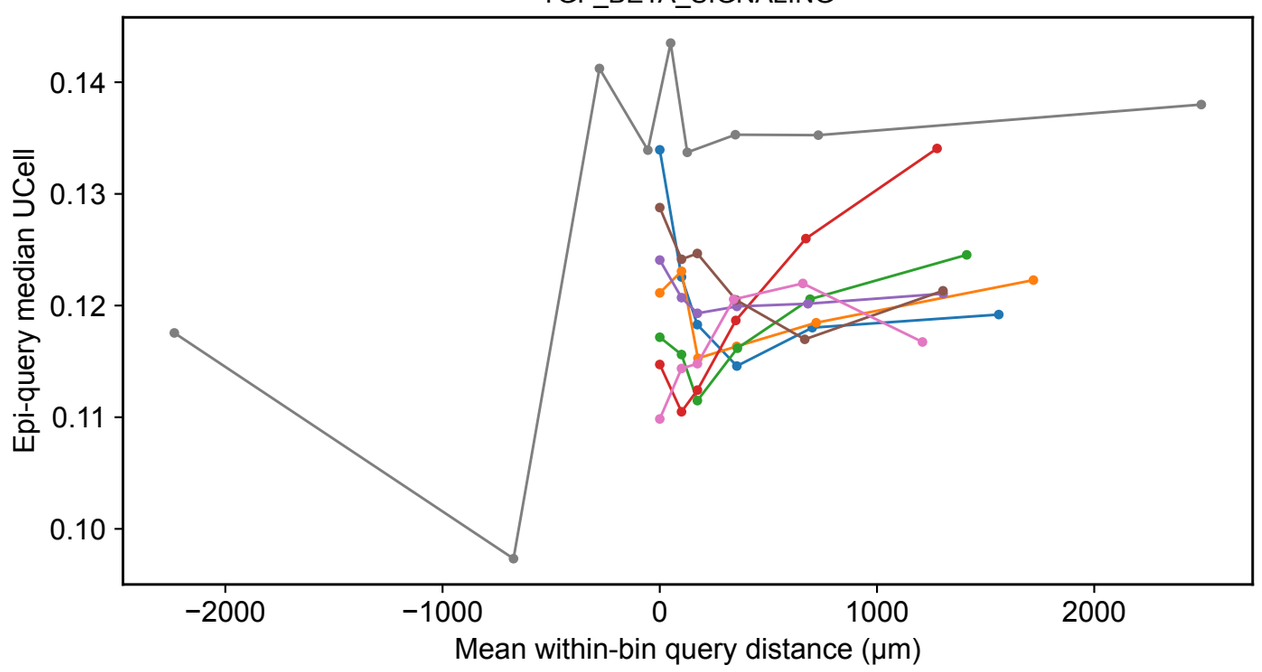
REACTIVE_OXYGEN_SPECIES_PATHWAY



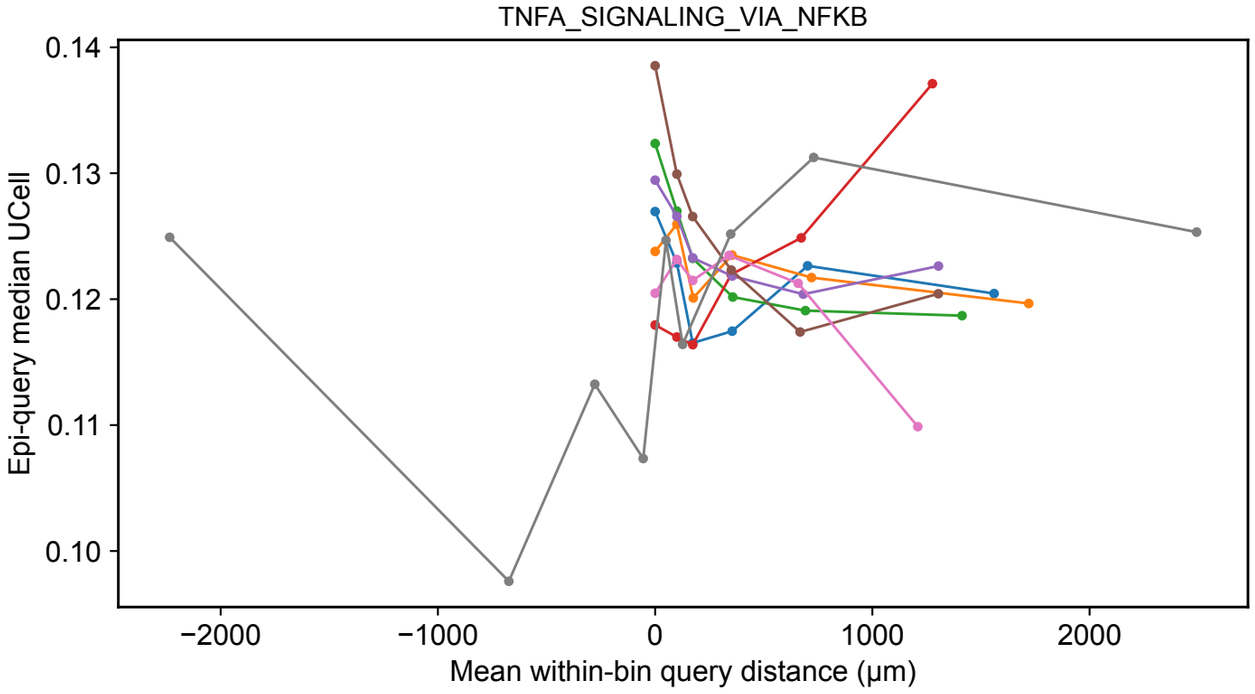
SPERMATOGENESIS



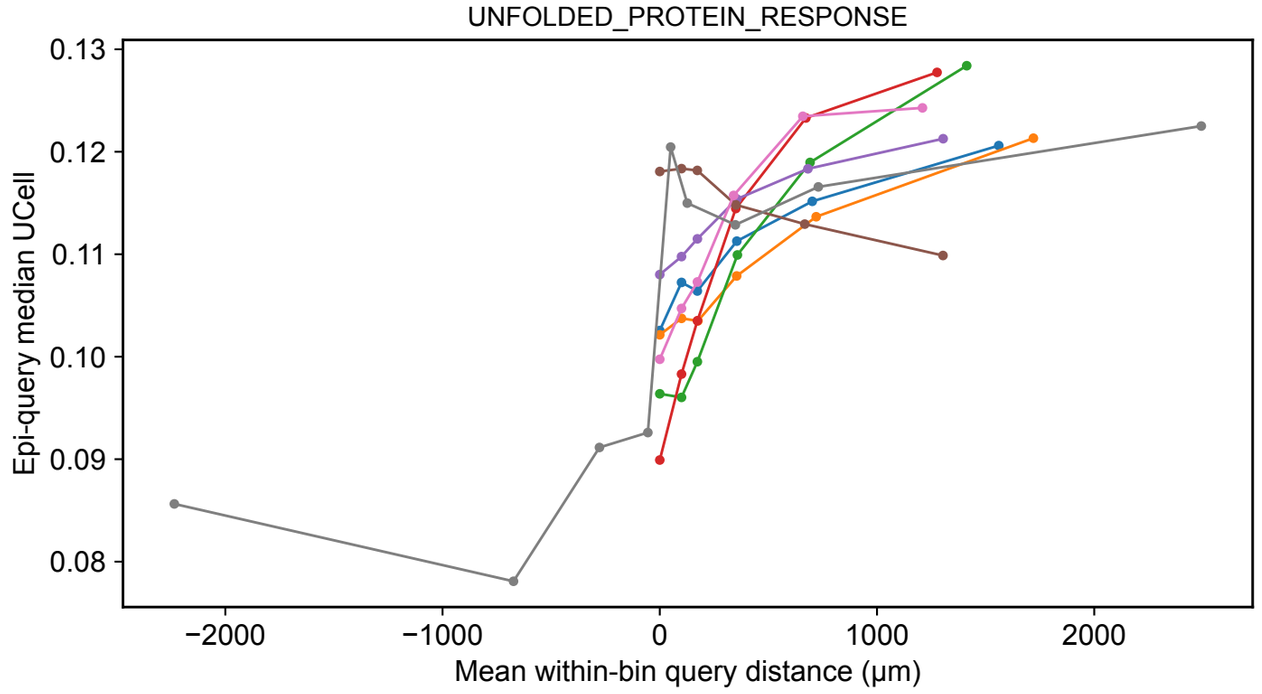
TGF_BETA_SIGNALING



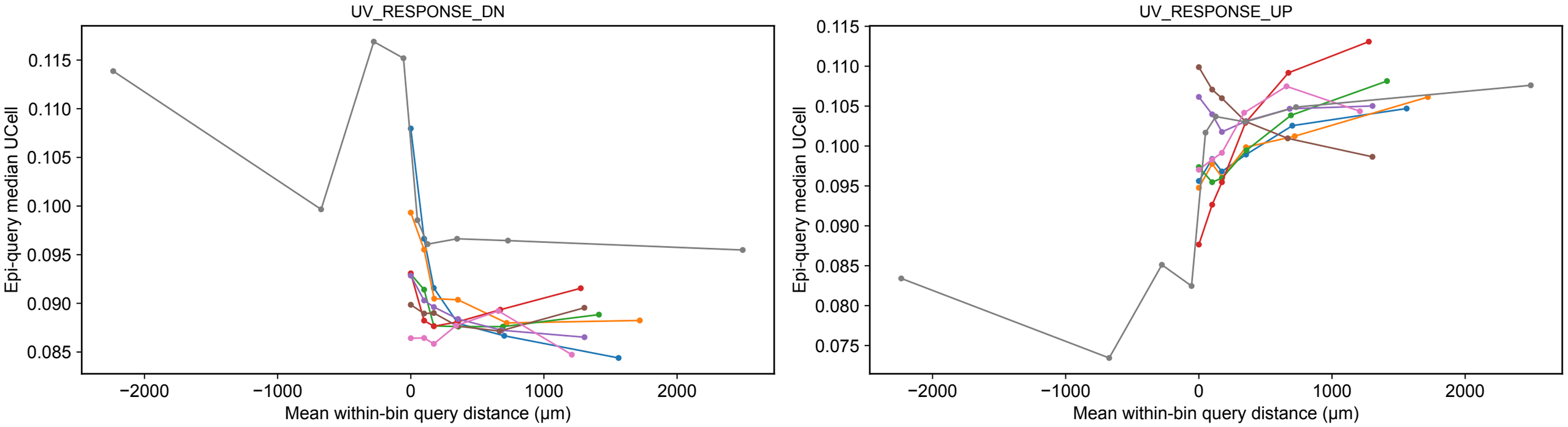
TNFA_SIGNALING_VIA_NFKB



UNFOLDED_PROTEIN_RESPONSE






UV_RESPONSE_DN
UV_RESPONSE_UP
WNT_BETA_CATENIN_SIGNALING
XENOBIOTIC_METABOLISM
CHOLESTEROL_BIOSYNTHESIS
CO_INHIBITION_BY_CTLA4
EXTRACELLULAR_MATRIX_ORGANIZATION
Epi-query median UCell
Mean within-bin query distance (μm)
Matrix_CAF_P95
iCAF_P95
rCAF_P95
vCAF_P95
C1QC TAM_P95
SPP1 TAM_P95
Treg_P95
local_interface